Mandates, Trade Structure, and Trade Agreements*

Malachy James Gavan† & Maria Ptashkina‡

October 2026

Abstract

Before trade negotiations begin, governments commit to a mandate, a schedule of concessions contingent on the eventual agreement. We show that the strategic choice of a mandate can prevent an efficient agreement even when both governments would gain from one and can compensate each other freely, and we show when it does: it turns on a single second-order property of the environment, the modularity of aggregate welfare with respect to tariffs. When strictly submodular, the governments' unilateral claims exceed the available surplus and no efficient agreement can be sustained; when strictly supermodular, an efficient agreement can instead be supported. A general-equilibrium foundation links this property to the structure of trade, where it depends on whether a country pair specialises within or across sectors, a distinction visible in trade data. Bilateral trade data provide illustrative evidence consistent with the mechanism. Holding a pair's overall balance of trade fixed, specialisation across rather than within sectors is associated with a higher probability of agreement. The mechanism is not special to tariffs, and the same distinction applies to when independently chosen contingent transfers can sustain efficient cooperation in other strategic settings.

JEL Codes: C72, C78, D23, F13

*We thank Paola Conconi, Pia Ennuschat, Alexey Gorn, Gene Grossman, Simon Loertscher, Phillip McCalman, Balazs Murakozy, Antonio Penta, Giacomo Ponzetto, Marek Pycia, Robert Staiger, and Jaume Ventura for their comments and suggestions. We thank Lia Palminteri for excellent research assistance. We thank the seminar participants at the University of Liverpool, the University of Melbourne, the University of Bologna, Universitat Pompeu Fabra, OASIS Trade Workshop, the University of Auckland, METC Conference, the Berlin School of Economics and Sabancı University for their comments.

†University of Liverpool: malachy@liverpool.ac.uk

‡University of Melbourne: maria.ptashkina@unimelb.edu.au

1 Introduction

Before a trade negotiation begins, each government sets a negotiating mandate: a statement of its objectives and of the concessions its negotiators are authorised to make. Because the mandate is fixed before bargaining and limits what can later be offered, its design is itself strategic. A government may, for example, commit in advance to concede less in order to secure more favourable terms. Because both governments reason this way, the mandates they set to strengthen their own positions also determine which agreements remain within reach. Can strategic mandate-setting, on its own, prevent two governments from reaching an agreement that both would prefer to no agreement, even when they can compensate one another freely? It can. We show that mandates chosen in each side's interest can render every efficient agreement unsustainable, give a simple characterisation of when they do, and take that prediction to the structure of trade observed in the data.

To study this, we adopt the endogenous mechanism-design framework of Jackson and Wilkie (2005), which lets players commit, before they act, to side-payments contingent on the actions eventually taken. A negotiating mandate is naturally a commitment of this kind. In our setting the actions are the two governments' tariffs, and a mandate is a schedule of side-payments fixed before tariffs are chosen and contingent on the tariff profile that emerges. These side-payments stand for the concessions governments exchange in real negotiations, such as market access or regulatory commitments, offered in exchange for the tariff outcome a government seeks.[1] A mandate might authorise a services-access concession only if the final tariff profile includes the agricultural cuts a country wants, so that what is offered depends on the negotiated outcome rather than being offered unconditionally.

We work with two-country environments carrying the usual gains from tariff cooper-

[1] Treating the concessions exchanged in negotiations as transfers is a standard modelling device; for example, Bagwell and Staiger (2010a) and Aghion et al. (2007) both represent the concessions governments make to one another as side-payments.

ation: each government is harmed by its partner's tariff, and non-cooperative tariffs are inefficiently high. Because concessions merely redistribute surplus, any inefficiency we find stems from how mandates are chosen; we accordingly set aside political constraints (Grossman and Helpman, 1995; Mitra, 2002), third-country and stumbling-block effects (Limão, 2006; Aghion et al., 2007; Karacaovali and Limão, 2008), costly contracting (Horn et al., 2010), and enforcement (Park, 2011; Buzard, 2017). Even so, cooperation can fail. That both governments prefer an agreement to the non-cooperative outcome does not make one sustainable: each must also be willing to write the mandate that delivers it. Taking its partner's mandate as given, a government can instead offer a different schedule of contingent concessions. An efficient agreement must therefore leave each government at least what such a unilateral choice can secure. When these two claims together exceed the surplus available from cooperation, no division satisfies both, and no efficient agreement is sustainable.

Our theoretical results provide a primitive classification of when strategic mandate choice is compatible with efficient agreement. The key property is the modularity of aggregate welfare in the two tariffs. Welfare is submodular when the gain from lowering one tariff is larger the higher the other, so that the two reductions are substitutes in generating surplus. Under strict submodularity, no efficient agreement can be sustained. Under strict supermodularity, the reductions are complements and, under additional regularity conditions, an efficient agreement can be sustained. Under submodularity the two unilateral claims together exhaust more than the available surplus; under supermodularity they leave enough room to support efficiency. Feasibility thus turns on the sign of a single cross-partial of aggregate welfare.[2] This is a second-order property, distinct from the familiar first-order observation that more complementary economies may gain more from trade. That observation concerns

[2]The two directions are not equally demanding. The impossibility result needs only that aggregate welfare be strictly submodular, that a pure-strategy Nash equilibrium exist, and that Nash tariffs be inefficiently high. The possibility result asks more, adding a strict terms-of-trade externality, joint concavity of aggregate welfare, two regularity conditions on individual payoffs (diminishing harm and stable individual modularity), and a sufficiently fine tariff grid. Section 4 gives the precise statements.

the level of the gains. Whether a mandate-sustained agreement exists turns instead on their curvature, an object that has received little attention in trade theory.

What, then, determines whether aggregate welfare is submodular or supermodular? To answer this, we provide a general-equilibrium foundation in which two countries trade differentiated goods under nested-CES preferences, with goods substitutes within sectors and complements across sectors, and governments maximising national welfare as in the standard terms-of-trade approach to trade agreements (Bagwell and Staiger, 1999). In this environment, the modularity of aggregate welfare depends on where the two countries' specialisation lies in the product hierarchy. Overlap within sectors pushes welfare towards submodularity, while specialisation across sectors pushes it towards supermodularity. What matters is therefore not simply how much the two countries trade, but the level of the product hierarchy at which their trade overlaps.

The mechanism operates through world-price adjustments. Consider first trade within a narrow product group, where goods are substitutes. When one country lowers its tariff, the good it imports becomes relatively cheaper at home and demand shifts towards it, so its world price rises to clear the world market. That same movement is felt abroad, where it makes the good imported by the partner relatively cheaper and shifts the partner's demand towards it. The tariff cut therefore does more than remove its own distortion: through the adjustment in world prices, it also offsets part of the distortion created by the partner's tariff. If the partner has already reduced its tariff, less of that distortion remains to offset, so the gain from cutting is smaller. The gain from one tariff reduction is therefore larger when the other tariff remains high, pushing aggregate welfare towards submodularity. When trade along this margin is sufficiently important to make aggregate welfare strictly submodular, our general result implies that no efficient agreement can be sustained. Across sectors of complements, the same world-price movement works in reverse. The induced rise in the

price of the good the partner exports depresses its demand for the complementary good it imports, so the gain from one tariff reduction is greater when the other tariff is also low, pushing aggregate welfare towards supermodularity.

We illustrate the model's implications using disaggregated bilateral trade and agreement data. For each country pair, we measure before any agreement is signed how much variation in trade direction occurs within sectors and how much occurs across them, and relate these measures to subsequent preferential-agreement formation. Between-sector variation is positively associated with agreement formation, while within-sector variation enters negatively. The difference between the two coefficients is positive and statistically significant: holding overall directional variation fixed, specialisation across rather than within sectors is associated with a higher signing probability. The magnitudes are economically meaningful. A one-standard-deviation change in the measures is associated with a difference equivalent to approximately 10–26 percent of the sample mean signing probability, depending on the measure and specification. Our empirical contribution is modest but focused: we ask whether prior trade structure is associated with agreement formation in the direction the model suggests, and we interpret the findings as illustrative evidence consistent with the mechanism rather than as a causal or structural test.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 introduces the formal model. Section 4 develops the main theoretical results. Section 5 provides the general-equilibrium foundation bridging theory and the data. Section 6 presents the empirical analysis. Section 7 concludes and shows that the same characterisation extends well beyond tariff-setting.

2 Related Literature

Our theoretical analysis builds on Jackson and Wilkie (2005), who study games in which players commit to action-contingent side payments before an underlying game is played. They characterise which action and payoff outcomes such commitments can sustain in equilibrium, through a condition on the most each player can guarantee by unilaterally redesigning its own commitment. This condition is general but implicit. For a structured class of bilateral environments, our contribution is to show that it reduces to a single, economically interpretable property, the modularity of aggregate payoff. This connects to work on contracting institutions and efficiency (Yamada, 2003; Ellingsen and Paltseva, 2016; Watson, 2024), transfers in networks (Bloch and Jackson, 2007), bargaining with externalities (de Fontenay and Gans, 2014), and side payments in political cooperation (Harstad, 2007). Modularity is central to comparative statics and strategic interaction (Topkis, 1998; Milgrom and Roberts, 1990; Milgrom and Shannon, 1994), matching (Becker, 1973), and oligopoly (Bulow et al., 1985). Here the relevant object is aggregate surplus, and the question is whether independently chosen commitments can sustain efficiency.

Our main economic setting is the theory of trade agreements. A large literature studies why governments may benefit from coordinating trade policy, including terms-of-trade externalities (Johnson, 1953; Bagwell and Staiger, 1999, 2002, 2010b), political-economy motives (Grossman and Helpman, 1994; Blanga-Gubbay et al., 2024), domestic-commitment motives (Maggi and Rodriguez-Clare, 1998; Maggi and Rodríguez-Clare, 2007), and uncertainty-management motives (Limão and Maggi, 2015); see Maggi (2014) and Grossman (2016) for reviews.

Negotiating institutions are a distinct topic in trade literature, and earlier work focused on preferential versus multilateral liberalisation (Bhagwati, 1992; Limão, 2006; Karacaovali and Limão, 2008), bargaining sequence and protocols (Aghion et al., 2007; Bagwell et al., 2020b;

Harstad, 2023), endogenous disagreement payoffs (Furusawa and Wen, 2002), and issue linkage (Conconi and Perroni, 2002; Limão, 2007; Maggi, 2016). Other contributions examine fast-track authority (Conconi et al., 2012), conditional concessions supporting domestic reforms (Conconi and Perroni, 2012) or conservation (Harstad, 2026), and self-enforcement under domestic credibility constraints, political incentives, or imperfect monitoring (Conconi and Perroni, 2009; Conconi and Sahuguet, 2009; Park, 2011; Buzard, 2017). We isolate a different obstacle: independently chosen mandates can prevent efficient agreements even without political frictions, costly contracting, or enforcement problems.

The structure of trade also has a well-established role in agreement formation. Production and specialisation patterns affect the formation and form of preferential agreements (Facchini et al., 2013, 2021), while trade structure can shape ratification incentives (Cole et al., 2021; Blanga-Gubbay et al., 2024) and the incentives generated by prospective future agreements (Lake, 2017). Foreign ownership and international production linkages also shape governments' tariff incentives (Blanchard, 2007; Blanchard et al., 2025). Our contribution is to show that agreements arise naturally under some trade structures and not others.

Finally, our empirical illustration complements evidence on economic, geographic, and political determinants of agreement formation (Mansfield et al., 2002; Magee, 2003; Baier and Bergstrand, 2004) and interdependence across agreements (Egger and Larch, 2008; Baldwin and Jaimovich, 2012); see Limão (2016). Our empirical exercise is motivated directly by the model's prediction that the location of similarity within the product hierarchy matters.

3 Model

This section sets out the model: a bilateral trade negotiation between two governments, conducted through the mandates each commits to before tariffs are chosen. We focus on two countries to capture the bilateral structure that characterises most preferential trade

negotiations.[3]

While our focus is the institutional setting of mandates in trade negotiations, the theoretical results are not special to that setting. In the conclusion we show that they hold in any environment satisfying the same assumptions in which contingent side-payments are available.

3.1 Tariff Payoffs

Consider a two-country model, $i = 1, 2$. Each country sets a tariff t_i, and we write $t = (t_1, t_2)$ for the tariff profile. The payoff of country i's government over tariff profiles is a continuous function $G_i : [0, \bar{t}]^2 \to \mathbb{R}_{++}$. Throughout, we fix the order of the arguments, writing $G_i(t_1, t_2)$, so that aggregate payoff is

$$G(t_1, t_2) = G_1(t_1, t_2) + G_2(t_1, t_2).$$

When it is convenient to place country i's own tariff first, we write $G(t_i; t_j)$ and $G_i(t_i; t_j)$, with the natural meaning. We have in mind two large trading countries whose tariff choices affect prices and, consequently, each other's economies and government payoffs as in Bagwell and Staiger (1999, 2002).

We impose a set of assumptions on G_1, G_2, and G that are standard in the literature on international trade agreements. The model is general enough to allow for alternative production structures, political economy considerations, and the presence of additional rest-of-the-world countries.[4] In later sections, we introduce a more structured trade environment

[3] In the DESTA dataset, around 60% of preferential agreements are purely bilateral, while a further 20% involve an existing trade bloc, typically bargaining with a common position, and another party. This differs from the multilateral setting, in which concessions are extended more broadly through the most-favoured-nation principle (Bagwell et al., 2020a).

[4] The reduced-form terms-of-trade payoff structure is consistent with a range of trade-agreement models, including alternative production and comparative-advantage structures (Bagwell and Staiger, 1999, 2002), political-economy weights on sectoral interests (Grossman and Helpman, 1994), and third-country or rest-

that allows us to relate the results below to the underlying structure of trade.

The first assumption is that a unilateral reduction in the tariff imposed by the other country is strictly beneficial to country i.

Assumption 1 (Negative externality)**.** *For each country i and each $t'_j > t_j$, $G_i(t_i; t_j) > G_i(t_i; t'_j)$.*

Let $t^{\text{eff}} = (t_1^{\text{eff}}, t_2^{\text{eff}}) \in \text{argmax}_{t \in [0,\bar{t}]^2} G(t)$ denote an aggregate-payoff-maximising tariff profile.

Assumption 2 (Excessive Nash tariffs)**.** *The tariff-setting game has at least one Nash equilibrium, and every Nash equilibrium $\hat{t}$ satisfies $\hat{t}_i > t_i^{\text{eff}}$ for $i = 1, 2$.*

We refer to this as *excessive Nash tariffs*: the equilibrium of the tariff-setting game involves tariffs higher than the efficient ones.

Assumption 3 (Joint concavity)**.** *The aggregate payoff function $G(t_1, t_2) = G_1(t_1, t_2) + G_2(t_1, t_2)$ is strictly jointly concave on $[0, \bar{t}]^2$.*

The third assumption is a curvature condition on aggregate payoff. These three assumptions are maintained throughout.

3.2 Concessions and Side-Payments

Following the institutional setting described earlier, the governments first outline a negotiating mandate which contains concessions contingent on the eventual tariff profile. To capture the concessions that governments make to one another during negotiations, we allow countries to make side-payments. Formally, country i chooses a side-payment schedule $S_i : [0, \bar{t}]^2 \to \mathbb{R}_+$, where $S_i(t)$ denotes the concession made by country i when the tariff profile

of-world extensions (Bagwell and Staiger, 2010b); see Maggi (2014) and Grossman (2016) for reviews.

is t. The restriction $S_i(t) \geq 0$ captures the idea that a country cannot unilaterally extract a concession from the other country; rather, concessions must be voluntarily offered.

Given a tariff profile t and a pair of side-payment schedules $S = (S_1, S_2)$, the total payoff of country i's government is

$$TG_i(t, S) = G_i(t) + S_{-i}(t) - S_i(t).$$

Side-payments only redistribute payoff between the two governments, so that $TG_1(t, S) + TG_2(t, S) = G(t)$. Side-payments enter payoffs linearly and separably from tariff payoffs. This abstracts from the direct welfare consequences of substantive concessions, such as provisions on intellectual property, labour standards, or regulation, and keeps the efficient tariff profile at the maximiser of aggregate payoff $G(t)$. Allowing such rich contingent compensation also removes incomplete contracting and limited transfers as explanations for failure, so any inefficiency arises from the strategic structure of mandate setting itself.

3.3 Timing and Equilibrium

We follow the endogenous mechanism design framework of Jackson and Wilkie (2005), which maps naturally to the institutional structure of predetermined mandates.

In the first stage, the two countries simultaneously choose their contingent side-payment schedules S_1 and S_2, their mandates, which are then observed by both, capturing the idea that once talks are under way each side understands the concessions the other is authorised to make as a function of the outcome.[5] In the second stage, countries simultaneously choose their tariffs. Each country retains unilateral control over its own tariff, taking as given the mandates chosen in the first stage and forming beliefs over the tariff its counterpart will

[5]One could instead leave residual uncertainty, strategic or structural, over the mandates; we do not, so as to isolate the inefficiency that arises from the institutional feature of mandate setting itself.

choose.

This timing places the model in a natural Coasean benchmark. If governments can compensate one another freely, it should not matter for efficiency which of them controls each tariff (Coase, 1960). What is different here is that this compensation is committed in advance, with each government choosing its own schedule. This still allows for reciprocity in the tariffs themselves, and not only in the transfers. By designing its side-payments appropriately, a government can make itself willing to lower its own tariff only when its counterpart does the same. The question is whether reciprocity of this kind, chosen independently and before tariffs are set, is enough to sustain an efficient agreement, which is itself reciprocal by definition.

Following Jackson and Wilkie (2005), we say that a tariff profile t together with a division u of $G(t)$ is supported if it is the outcome of a subgame perfect equilibrium (SPE). We allow mixing in the final tariff-choice stage but require the first-stage schedules to be chosen deterministically.[6] When no SPE with a deterministic outcome exists, we take it that no negotiation takes place and a Nash tariff profile, t^*, is the outcome, with payoffs $G_1(t^*)$ and $G_2(t^*)$.

Tariffs are chosen from a finite grid $T_\Delta \subset [0, \bar{t}]$ rather than the whole continuum $[0, \bar{t}]^2$. Finiteness of the action space guarantees that the final stage has a (possibly mixed) Nash equilibrium whatever schedules are chosen, which is what makes SPE well defined.[7] The grid

[6] Restricting attention to deterministic tariff outcomes is without loss for the efficiency question. By strict joint concavity of G (Assumption 3), t^{eff} is the unique maximiser of aggregate payoff, so any mixed profile placing positive probability on a tariff other than t^{eff} yields $\mathbb{E}_\mu[G(t)] < G(t^{\text{eff}})$ and is strictly inefficient. The relevant question is therefore whether t^{eff} is itself the on-path outcome of an equilibrium, with randomisation confined to off-path continuation play, rather than whether some fully mixed equilibrium attains the efficient payoff on average.

[7] Without a grid, existence is not automatic. If arbitrary schedules $S_i : [0, \bar{t}]^2 \to \mathbb{R}_+$ are permitted, the induced second-stage game need not possess a Nash equilibrium, even under the assumptions on G_1, G_2 that usually guarantee one, since a schedule can render a country's second-stage payoff discontinuous in the tariff profile. Existence is recovered if schedules are restricted to be continuous, and takes on the same form as the modified solo bounds (see the appendix of Jackson and Wilkie (2005)), and our results also extend to that case; we work on a grid because it delivers the same conclusions with substantially less technical overhead.

can be taken to be arbitrarily fine. We write $m(T_\Delta)$ for its mesh, the largest gap between adjacent grid points, and call a grid admissible if it contains $t_1^{eff}, t_2^{eff}, t_1^*$ and t_2^*.

For a mixed profile $\mu = \mu_1 \times \mu_2$, where $\mu_i \in \Delta(T_\Delta)$ and schedules are defined as mappings from points on the grid to payments $S_i : T_\Delta^2 \to \mathbb{R}_+$,[8] let the expected payoff of country i be denoted by

$$U_i(\mu, S) := \mathbb{E}_\mu[TG_i(t, S)] = \mathbb{E}_\mu[G_i(t) + S_j(t) - S_i(t)].$$

Note that $U_1(\mu, S) + U_2(\mu, S) = \mathbb{E}_\mu[G(t)]$ for all μ and all S. That is, regardless of mixing, side-payments are still purely redistributive. For a given pair of side-payment schedules S_i, S_j, let $NE(S_i, S_j)$ denote the set of (mixed) Nash equilibria in the final stage.

Before the formal apparatus, we outline the economics of the endogenous mechanism design framework in our setting. Mutual gains from tariff reductions are not enough to support an agreement: each government must also be willing to choose the mandate that delivers it. Taking its partner's mandate as given, a government could instead offer different concessions, changing the incentives that both countries face when setting tariffs. The question is whether this alternative allows it to secure a higher payoff. A revised mandate may, however, generate several tariff-setting equilibria. Because we ask whether an agreement can be supported in some subgame-perfect equilibrium, a less favourable continuation can deter a deviation, provided that continuation is itself an equilibrium. Conversely, if a revised mandate makes the government better off in every continuation equilibrium, no credible response can deter it. Jackson and Wilkie (2005) formalise this comparison through the "modified solo payoff": the highest payoff a government can secure by choosing its own mandate. An agreement must give each government at least this amount.

[8] Note that it is without loss to define S_i on either $[0, \bar{t}]^2$ or T_Δ^2, as ultimately only the side-payments on the grid are feasible.

3.4 Inherited Results

Jackson and Wilkie (2005) provide a complete characterisation, for general two-player finite action games, for when a tariff (action) and payoff pair can be sustained as the outcome of an SPE.[9] They consider a target profile of tariffs $\hat{t} = (\hat{t}_i, \hat{t}_j)$ and a division of the aggregate payoff $G(\hat{t})$, denoted by $u = (u_1, u_2)$ such that $u_1 + u_2 = G(\hat{t})$. They provide a minimal supporting schedule, which is the smallest non-negative schedule to ensure that country i's payoff is capped at u_i when the other is playing their targeted tariff, $\hat{t}_j$, while paying nothing elsewhere:

$$S_i^{\hat{t},u}(t_i, t_j) = \begin{cases} \max\{G_i(t_i; t_j) - u_i, 0\} & \text{if } t_j = \hat{t}_j \\ 0 & \text{if } t_j \neq \hat{t}_j \end{cases}$$

Their Theorem 3 then characterises exactly when an agreement stands. Write $u_i^{ms}(t, u)$ for country i's "modified solo payoff": the most i can obtain by unilaterally redesigning its own mandate, holding its partner's minimal supporting schedule fixed, when the continuation that follows is the one least favourable to i. A tariff and payoff pair (t, u) is the outcome of a subgame perfect equilibrium if and only if it concedes each country at least this value,

$$u_i \geq u_i^{ms}(t, u) = \sup_{S_i'} \min_{\mu \in NE(S_i', S_j^{t,u})} U_i(\mu, (S_i', S_j^{t,u})).$$

Two features of this criterion are worth emphasising. The supremum over S_i' makes it a statement about all unilateral mandate deviations. The minimum over continuations makes it demanding of the deviator rather than of the agreement: a deviation counts only if it pays even under the continuation the deviator likes least.

The limitation of the above result for our purposes is that the condition $u_i \geq u_i^{ms}$ is

[9] Note in Jackson and Wilkie (2005), t is used to denote side-payments (transfers), and x to denote actions (tariffs).

implicit. The modified solo payoff u_i^{ms} is defined by a nested optimisation over alternative mandates and the continuation equilibria they induce, so evaluating it requires solving the problem for the particular payoff functions at hand. It therefore does not by itself provide a primitive classification of which environments can sustain an efficient outcome.

4 Equilibrium Outcomes and Modularity

In this section, we show that whether an efficient agreement can be sustained is organised around a single property of aggregate payoff, its modularity, read directly from the environment, rather than relying on the nested characterisation of Jackson and Wilkie (2005). Strict submodularity rules an efficient agreement out, while strict supermodularity supports one, under the maintained assumptions and two further regularity conditions introduced below.

The argument runs through a tractable lower bound on each country's modified solo payoff, obtained from unilateral mandate deviations that induce a pure continuation. We call it the country's pure solo claim. If the two pure solo claims together exceed the surplus available from an efficient agreement, the modified solo payoffs must exceed it as well, no division satisfies both countries, and no efficient agreement is sustainable. Strict submodularity of aggregate welfare is exactly what makes the claims overshoot in this way, which is why the impossibility result needs little beyond it. The positive direction is more delicate, and is relegated mostly to the appendix. Under supermodularity the claims instead fall short of the efficient surplus, but this alone does not establish that an agreement can be sustained, and one must also control deviations whose continuation is mixed, and hence the full modified solo payoff.

4.1 Pure Solo Claims

Suppose country i abandons whatever mandate it was going to write and instead designs a schedule with a single target tariff profile in mind. It can do three things at once. Firstly, it can return the concessions its partner has offered, so that the partner's mandate confers no advantage. Secondly, it can pay the partner exactly enough to accept the tariff i wants it to set. Finally, it can commit itself to its own target by promising large payments at every other own tariff. As that commitment works, it has no bite on the path. What is left over is the aggregate payoff at the target, less whatever the partner could have secured on its own. Formally, for a tariff $t_i \in [0, \bar{t}]$,

$$\underline{TG}_i(t_i) := \underbrace{\max_{t_j \in T_\Delta} G(t_i; t_j)}_{\text{best aggregate payoff at } t_i} - \underbrace{\max_{t'_j \in T_\Delta} G_j(t_i; t'_j)}_{\text{what the partner can secure}} .$$

Given that this can be done for any grid tariff, we define the "pure solo claim"[10] as

$$\underline{TG}_i := \max_{t_i \in T_\Delta} \underline{TG}_i(t_i).$$

The claim $\underline{TG}_i$ is a conservative estimate of what a deviation is worth. Against a partner running a schedule S_j, country i need only leave the partner its payoff net of its own payments, so the partner term $\max_{t'_j} G_j(t_i; t'_j)$ falls to $\max_{t'_j}[G_j(t_i; t'_j) - S_j(t_i; t'_j)]$. A partner's own concessions lower what it can secure and so make it cheaper to buy off. This observation returns later, where it becomes the central obstacle.

The construction as described points towards inducing indifference by paying the partner exactly enough to ensure the target is *one* of the best responses, and does not readily rule

[10]Both maxima run over the grid T_Δ, so the claim is a genuine feasible-deviation value. The positive result of Section 4.3 works instead with a *hybrid* claim $\widehat{TG}_i$ (Appendix B): the deviator's own tariff and the aggregate maximisation range over $[0, \bar{t}]$, to accommodate the mean of a mixed continuation and to keep the aggregate term concave, while the partner's response is held to the grid.

out other equilibria in the continuation, and therefore may not be a true lower bound. The following lemma removes the issue by exhibiting a deviation whose continuation is unique.

Lemma 1. *$u_i^{ms}(\hat{t}, u) \geq \underline{TG}_i$ for every target $\hat{t}$ and division u. Therefore any outcome must guarantee at least $\underline{TG}_i$ for each country.*

The proof is in Appendix A. We note that this does not depend on any assumptions made about G_1, G_2, or G.

4.2 Impossibility under Submodularity

With the definition of the pure solo claim in hand, and knowing that in any equilibrium each country must receive at least its pure solo claim, we can now turn to our first result connecting this to the modularity of the aggregate payoff.

To do so, let us recall the definition of strict submodularity. We say that G is strictly submodular if, for any $t_1 > t_1'$ and $t_2 > t_2'$,

$$\underbrace{G(t_1', t_2') - G(t_1, t_2')}_{\substack{\text{aggregate gain from a tariff reduction} \\ \text{when the other tariff is low}}} < \underbrace{G(t_1', t_2) - G(t_1, t_2)}_{\substack{\text{aggregate gain from a tariff reduction} \\ \text{when the other tariff is high}}} .$$

Strict submodularity implies that the aggregate payoff gain from a reduction in one country's tariff is greater when the other country's tariff is higher. In the environment considered here, where efficient tariffs are lower than the Nash tariffs, this captures the idea that tariff reductions become more valuable when the countries' tariff choices are mutually further from the low-tariff outcome.

When G is twice differentiable, a useful characterisation of strict submodularity is $\frac{\partial^2 G}{\partial t_1 \partial t_2} < 0$. Importantly, strict submodularity of the aggregate payoff function does not require either

individual government's payoff function to be strictly submodular. Conversely, if both G_1 and G_2 are strictly submodular in the two tariffs, then their aggregate, G, is also strictly submodular.

We can now state the first main result.

Proposition 1. *If the aggregate payoff G is strictly submodular, then $\underline{TG}_1 + \underline{TG}_2 > G(t^{eff})$. No pure tariff outcome is then supportable in the mandate game, so the induced profile is the non-cooperative t^*, with payoffs $(G_1(t^*), G_2(t^*))$.*

The logic is the demand argument of the introduction, in terms of the pure solo claim. By Lemma 1, any agreement must deliver each country at least its claim $\underline{TG}_i$, and under strict submodularity that claim is large. Evaluate country i's claim at its own non-cooperative tariff t_i^*: the partner's best reply is t_j^*, so the most it can secure alone is $G_j(t^*)$, while moving it to the efficient tariff t_j^{eff} makes $G(t_i^*; t_j^{eff})$ available, so that

$$\underline{TG}_i \;\geq\; G(t_i^*; t_j^{eff}) - G_j(t^*).$$

Both countries can make this demand at once. Summing, and using strict submodularity to compare the efficient profile t^{eff} with the uniformly higher t^*, gives $\underline{TG}_1 + \underline{TG}_2 > G(t^{eff})$: the two claims exceed the entire efficient surplus (Figure 1), so no division satisfies both, and the outcome reverts to the non-cooperative t^*, which exists by assumption. The formal proof is in Appendix A.

Proposition 1 is notably undemanding. It does not rely on joint concavity, on the individual-payoff conditions, or on a tariff reduction by the partner being beneficial. Beyond strict submodularity, it requires only the existence of a pure Nash equilibrium, assumed to lie on the grid, and excessive Nash tariffs.

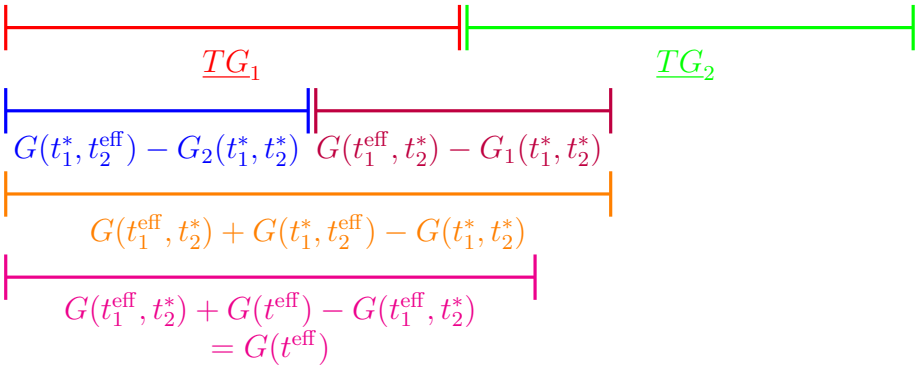


Figure 1: Impossibility under strict submodularity. Each country's pure solo claim $\underline{TG}_i$ (top two bars) is at least what it can secure by conceding the efficient tariff to its partner while keeping its own non-cooperative tariff: the blue and purple bars are these lower bounds for countries 1 and 2. Strict submodularity makes their sum (orange) exceed the efficient surplus $G(t^{\text{eff}})$ (magenta), so the two claims cannot both be met and no efficient agreement is supportable.

4.3 Possibility under Supermodularity

We now turn to the positive case, in which the aggregate payoff is strictly supermodular: for $t_1' < t_1$ and $t_2' < t_2$,

$$\underbrace{G(t_1', t_2') - G(t_1, t_2')}_{\substack{\text{aggregate gain from a tariff reduction} \\ \text{when the other tariff is low}}} > \underbrace{G(t_1', t_2) - G(t_1, t_2)}_{\substack{\text{aggregate gain from a tariff reduction} \\ \text{when the other tariff is high}}}.$$

The aggregate gain from one country's tariff reduction is now larger when the other's tariff is also low, so the two reductions are complements; when G is twice differentiable this is $\frac{\partial^2 G}{\partial t_1 \partial t_2} > 0$.

Strict supermodularity reverses the comparison in Proposition 1. Weighed against the same profiles, the two countries' pure solo claims now fall short of the efficient surplus,

$$\underline{TG}_1 + \underline{TG}_2 \;<\; G(t^{eff}),$$

leaving strictly positive room beneath it (Lemma 5 in Appendix B.3). This strict slack is what allows an efficient agreement to be supported. For this result we require two further conditions, both on individual rather than aggregate payoffs.

Assumption 4 (Diminishing harm)**.** *For each country i, $G_i(t_i; t_j)$ is weakly convex in t_j.*

Assumption 5 (Stable individual modularity)**.** *For each country i, the payoff G_i is either weakly supermodular throughout $[0, \bar{t}]^2$ or weakly submodular throughout $[0, \bar{t}]^2$.*

Both matter only for the positive result. Diminishing harm ensures a government cannot profit by making its tariff unpredictable instead of committing to the efficient level, since spreading it out does not lower what its partner can secure on its own. Stable individual modularity ensures each country's best reply is monotone in its partner's tariff.

Stable individual modularity does not require tariffs to be strictly supermodular, and therefore strategic complements. The direction of the cross-effect may even differ across the two countries: one country's tariffs may be strategic complements while the other's are substitutes (Bulow et al., 1985), though not both substitutes, since their sum, the aggregate, is supermodular. What it rules out is a change in the sign of the strategic interaction within a country. Neither condition constrains aggregate modularity, which remains the object that governs the economics.

We can now state the possibility result.

Proposition 2 (Efficient agreement under supermodularity)**.** *Suppose G is strictly supermodular, and that diminishing harm and stable individual modularity hold. Then there exists $\bar{\Delta} > 0$ such that for every admissible tariff grid with mesh $m(T_\Delta) < \bar{\Delta}$, there is a subgame perfect equilibrium implementing t^{eff}.*

The construction is in Appendix B, and runs by an analogous logic as the impossibility result. The one new difficulty is that the bound on a deviation must be constructed while the

partner holds its minimal supporting schedule, which requires care once a deviation induces a mixed continuation; we defer the details to the appendix.

Remark 1 (The modular boundary)**.** *At the boundary between the two cases, aggregate payoff is exactly modular and the aggregate gain from one country's tariff reduction is the same whether or not the other has already reduced its own. There is then no strict slack, and aggregate modularity alone no longer suffices. Supporting* t^{eff} *requires the stronger condition that each individual payoff be modular. We treat this case separately in Appendix C.*

5 Relation to Trade Structure through a GE Foundation of Payoffs

Section 4 reduced the feasibility of an efficient agreement to the modularity of aggregate welfare. This section asks what determines that modularity in a trading economy. We build a general-equilibrium foundation in which the answer lies in where the two countries' specialisation sits in the product hierarchy: specialisation in different goods within the same sector pushes aggregate welfare towards submodularity, and specialisation across sectors pushes it towards supermodularity. This yields an empirical prediction that we take to the data in Section 6.

5.1 A Nested CES Endowment Economy

We consider an endowment economy with two countries, K sub-sectors, and L differentiated goods within each sub-sector. Each country is represented by a representative consumer who owns the country's endowments. A differentiated good is indexed by $g = (k, l)$. Preferences take a two-level nested CES form, with within-sub-sector elasticity $\sigma_k > 1$ and across-sub-sector elasticity $\sigma_0 < 1$. The second inequality is the stronger, substantive assumption: broad product groups are taken to be complements, not merely weaker substitutes. In addition, there is a freely traded numeraire good that enters utility linearly.

Specifically, let x_i denote consumption of the numeraire and $c_i = (c_{i,k,l})_{k,l}$ consumption of the differentiated goods by the representative consumer in country i. Utility is

$$U_i(x_i, c_i) = x_i + \gamma_i \log V(c_i),$$

where

$$V(c_i) = \Big[\sum_k \alpha_k Q_{i,k}^{\frac{\sigma_0-1}{\sigma_0}}\Big]^{\frac{\sigma_0}{\sigma_0-1}}, \qquad Q_{i,k} = \Big[\sum_l \beta_{kl} c_{i,k,l}^{\frac{\sigma_k-1}{\sigma_k}}\Big]^{\frac{\sigma_k}{\sigma_k-1}}.$$

Here, $\gamma_i > 0$ determines country i's expenditure on differentiated goods, while α_k and β_{kl} are respectively the across- and within-sub-sector CES weights.

Note that the side-payments of the general model can be interpreted directly as transfers of the numeraire good.

Endowments. Country i is endowed with $\bar{x}_i$ units of the numeraire good and $e_{i,k,l} \geq 0$ units of each differentiated good (k, l). We assume that the endowment of the numeraire is sufficiently large that numeraire consumption remains positive for all tariff profiles considered. The numeraire therefore absorbs wealth effects and pins down the price level, so demand for differentiated goods is independent of income. Differences in the vectors $(e_{i,k,l})_{k,l}$ determine the countries' patterns of specialisation and, ultimately, the structure of trade between them.

Demand. Let $p_{i,k,l}$ denote the domestic price faced by country i for good (k, l). Given these prices, the CES price index for sub-sector k is $R_{i,k} = \big[\sum_l \beta_{kl}^{\sigma_k} p_{i,k,l}^{1-\sigma_k}\big]^{\frac{1}{1-\sigma_k}}$, while the corresponding price index for the aggregate of differentiated goods is $P_i = \big[\sum_k \alpha_k^{\sigma_0} R_{i,k}^{1-\sigma_0}\big]^{\frac{1}{1-\sigma_0}}$. The nested CES structure implies that the expenditure share allocated to sub-sector k

is $\theta_{i,k} = \alpha_k^{\sigma_0} \left(\frac{R_{i,k}}{P_i} \right)^{1-\sigma_0}$, while the share of expenditure within sub-sector k allocated to good (k,l) is $s_{i,k,l} = \beta_{kl}^{\sigma_k} \left(\frac{p_{i,k,l}}{R_{i,k}} \right)^{1-\sigma_k}$. Since the logarithmic specification implies that country i spends exactly γ_i on differentiated goods, demand for good (k,l) is therefore $c_{i,k,l} = \frac{\gamma_i \theta_{i,k} s_{i,k,l}}{p_{i,k,l}}$.

World Prices and Market Clearing. Let $p_{k,l}^w$ denote the world price of good (k,l), with the price of the freely traded numeraire normalised to one. We work locally around free trade and hold the direction of trade fixed. Let $m_{i,k,l} \in \{0,1\}$ indicate whether country i imports good (k,l), where $m_{1,k,l} + m_{2,k,l} = 1$ for every good. Country i imposes a uniform tariff on the differentiated goods that it imports. Writing t_i for the log gross tariff, the domestic price faced by country i is therefore $p_{i,k,l} = e^{m_{i,k,l} t_i} p_{k,l}^w$: an importing country faces the tariff-inclusive price $e^{t_i} p_{k,l}^w$, while an exporting country faces the world price $p_{k,l}^w$. Given these domestic prices and the demands described above, the equilibrium world price of each good is determined by the market-clearing condition $c_{1,k,l} + c_{2,k,l} = e_{1,k,l} + e_{2,k,l}$.

At free trade, the relative endowments of each good determine which country is the importer, so the vector $(m_{i,k,l})_{k,l}$ is itself pinned down by the countries' endowment structure, which we assume does not change depending on tariffs.

Income. Country i receives income from its endowment of the numeraire, the value of its differentiated goods endowments, and tariff revenue. Letting TR_i denote tariff revenue, country i's income is therefore $Y_i = \bar{x}_i + \sum_{k,l} p_{i,k,l} e_{i,k,l} + TR_i$, where $TR_i = (e^{t_i} - 1) \sum_{k,l} m_{i,k,l} p_{k,l}^w (c_{i,k,l} - e_{i,k,l})$. The representative consumer faces the budget constraint $x_i + \sum_{k,l} p_{i,k,l} c_{i,k,l} \leq Y_i$. Since utility is quasi-linear in the numeraire, changes in income are absorbed through x_i and do not affect demand for differentiated goods.

Welfare. Government i evaluates the welfare of its representative consumer. Using the budget constraint and the fact that expenditure on differentiated goods is equal to γ_i, indirect utility can be written as $G_i(t_i, t_j) = Y_i - \gamma_i + \gamma_i \log \gamma_i - \gamma_i \log P_i$. Aggregate welfare is therefore $G(t_1, t_2) = G_1(t_1, t_2) + G_2(t_1, t_2)$. Using market clearing for all goods, income and tariff-revenue terms cancel in aggregate, so that $G(t_1, t_2) = \bar{x}_1 + \bar{x}_2 + \sum_i \gamma_i \log \gamma_i - \sum_i \gamma_i \log P_i$. Hence, for the purposes of the modularity results below, tariffs affect aggregate welfare through their effect on the two countries' differentiated-good price indices.

Relation to the General Framework. Appendix D states additional local regularity conditions on the world-price response to tariffs and shows that the government payoffs generated by this economy satisfy all of the maintained assumptions of the general theoretical results. In this environment the efficient tariff profile is free trade, $t^{eff} = (0, 0)$: tariffs enter only as consumption distortions, so aggregate welfare is maximised when both are zero, while the excessive-Nash property places the non-cooperative tariffs t^* strictly above it. Throughout, we take the upper bound on tariffs, $\bar{t}$, to be "small". The relevant properties of the government payoff functions can therefore be established locally around free trade and, by continuity, hold throughout the tariff domain $[0, \bar{t}]^2$.

5.2 Trade Structure and Modularity

We now characterise how the location of trade within the nested product structure determines the modularity of aggregate welfare. At free trade, both countries face the same differentiated-good prices and, since preferences are identical, have common expenditure shares. We therefore write $s_{k,l}$ for the expenditure share of good (k, l) within sub-sector k and θ_k for the expenditure share of sub-sector k. For notational simplicity, let $m_{k,l} = m_{1,k,l}$ denote whether country 1 imports good (k, l), so that country 2 imports the good whenever $m_{k,l} = 0$. Define

$$\bar{m}_k = \sum_l s_{k,l} m_{k,l} \quad \text{and} \quad \text{Var}^\theta(\bar{m}_k) = \sum_k \theta_k \bar{m}_k^2 - \left(\sum_k \theta_k \bar{m}_k \right)^2 .$$

The first object is the expenditure share of goods within sub-sector k that are imported by country 1, while the second is the expenditure-weighted variance of this share across sub-sectors.[11]

These two objects, $\bar{m}_k(1-\bar{m}_k)$ and $\text{Var}^\theta(\bar{m}_k)$, capture distinct patterns of specialisation between the countries. If the countries specialise in different goods within the same sub-sector, importer identity varies within that sub-sector and $\bar{m}_k(1-\bar{m}_k)$ is large. By contrast, if each country tends to specialise in different sub-sectors, then $\bar{m}_k$ will be close to either zero or one within each sub-sector but will vary substantially across k, making $\text{Var}^\theta(\bar{m}_k)$ large. Hence, two country pairs can have a similar overall degree of trade while differing substantially in whether their specialisation occurs within or across sectors. The proposition below shows that it is precisely this location of specialisation within the product structure that determines the modularity of aggregate welfare.

Proposition 3 (Trade Structure and Modularity)**.** *At free trade,*

$$\frac{\partial^2 G}{\partial t_1 \partial t_2} = \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \left[\sum_k \theta_k (1-\sigma_k) \bar{m}_k (1-\bar{m}_k) + (1-\sigma_0) \, \text{Var}^\theta(\bar{m}_k) \right].$$

The derivation is in Appendix D. The result separates trade structure into within- and between-sector components. When the countries specialise in different goods within the same sub-sector, the first term becomes more important; since $\sigma_k > 1$ it is negative, pushing aggregate welfare towards submodularity. When instead the countries specialise in different sub-sectors, the second term becomes more important; since $\sigma_0 < 1$ it is positive, pushing

[11] Recall that $\theta_k = \alpha_k^{\sigma_0} \left(\frac{R_k}{P} \right)^{1-\sigma_0}$, $s_{k,l} = \beta_{kl}^{\sigma_k} \left(\frac{p_{k,l}}{R_k} \right)^{1-\sigma_k}$, $R_k = \left[\sum_l \beta_{kl}^{\sigma_k} p_{k,l}^{1-\sigma_k} \right]^{\frac{1}{1-\sigma_k}}$ and $P = \left[\sum_k \alpha_k^{\sigma_0} R_k^{1-\sigma_0} \right]^{\frac{1}{1-\sigma_0}}$, where, at free trade, the country index i is suppressed.

aggregate welfare towards supermodularity. Thus, trade between countries that compete within narrowly defined products makes agreement more difficult to sustain, while trade based on specialisation in different, more complementary product groups makes agreement easier to sustain.

6 Empirical Evidence

Proposition 3 distinguishes two patterns of bilateral specialisation, trade in different goods within the same sectors and specialisation across sectors, the first pushing aggregate welfare toward submodularity and the second toward supermodularity. We ask whether these features of a pair's prior trade structure are associated with the subsequent formation of a preferential agreement. Given the stylised nature of the model, we read the exercise as descriptive evidence on the mechanism rather than a structural or causal test of it.

6.1 Measuring Within- and Between-Sector Trade Structure

To empirically approximate the theoretical objects in Proposition 3 we use the structure of the Harmonised System. The HS2 chapters correspond to sectors k, and the within-sector trade variation is defined at HS6 level.

For a given country-pair i, j and year, let $x_{ij,g}$ denote exports of product g from i to j. Suppressing the year index, define:

$$v_g = x_{ij,g} + x_{ji,g}, \quad q_g = \frac{x_{ji,g}}{v_g}, \quad \omega_g = \frac{v_g}{\sum_h v_h},$$

for products with positive bilateral trade. Here, q_g measures the trade direction, i.e. the fraction of the product's bilateral trade imported by country i.[12] The weight ω_g measures

[12] In theory we assume net importers, but in the data there are products that flow both ways.

the product's importance in the bilateral trade basket.

For each sector k, define:

$$\theta_k = \sum_{g \in k} \omega_g, \quad q_k = \frac{\sum_{g \in k} \omega_g q_g}{\theta_k}, \quad \mu = \sum_k \theta_k q_k,$$

where θ_k tells us how important each sector k is, and the sector weights sum up to one. q_k is the weighted average of the product-level directions, and μ is the average import direction across all bilateral trade.

With these definitions at hand, we can construct the within- and between-sector measures as:

$$W_{ij,t} = \sum_k \theta_k q_k (1 - q_k), \quad B_{ij,t} = \sum_k \theta_k (q_k - \mu)^2.$$

The within-sector measure W shows how much two-way trade occurs within sectors. To gain intuition, note that $q_k \in [0, 1]$, so the expression is zero when one country is completely specialised in a sector, and reaches its maximum when the sector trade is perfectly balanced. Importantly, W does not only measure trade in the same HS6 goods. Suppose a HS2 sector contains cars and car parts, and country i exports cars and imports car parts. The HS2 sector as a whole has two-way trade which contributes to W.

The between-sector measure B compares the direction of trade in each sector, q_k, with the direction of trade for the bilateral relationship as a whole, μ. It shows how much the direction of trade differs across sectors. B is high when country i mainly imports some sectors and exports others.

These measures represent a classic decomposition of the overall variation:

$$H_{ij,t} \equiv W_{ij,t} + B_{ij,t} = \mu(1 - \mu).$$

Two pairs can have the same overall directional balance but differ in where the two directions of trade occur. This comparison isolates the location of bilateral specialisation rather than the volume of trade. Thus, we can ask: holding the total amount of directional variation H fixed, how does agreement formation differ when that variation occurs between sectors rather than within sectors?

These measures approximate, rather than completely recover, the theoretical objects in Proposition 3. Bilateral trade shares replace the model's expenditure shares, HS2 chapters approximate its sectors, and observed two-way trade within an HS6 category reflects finer varieties traded in opposite directions.

6.2 Data and Empirical Specification

To construct the within- and between-sector measures we use UN Comtrade data for 2002–2014.[13] Agreement histories come from DESTA. The outcome is the first signature of a full free trade agreement or customs union for which an entry-into-force year is recorded. For each outcome year t, we measure prior trade structure using the average of the annual measures in years $t-7, t-6, t-5$. We denote these averages by $\bar{W}_{ij,t}$ and $\bar{B}_{ij,t}$. This window describes trade structure several years before signature, and averages trade flows to avoid lumpiness. The estimating sample contains 34,830 pair-year observations for 5,587 pairs and 159 countries or territories. Since we are interested in how trade structure prior to agreement formation is associated with the probability of agreement formation, we restrict the agreements to a period over 2009–2019. It includes 521 first-signature observations. Geographic and GDP controls come from CEPII.

We estimate a linear probability model:

[13] These years maintain the same HS2002 classification, and avoid classification changes. See the full description of the data construction in Appendix E.

$$\text{PTA}_{ij,t} = \beta_W \bar{W}_{ij,t} + \beta_B \bar{B}_{ij,t} + X'_{ij,t}\gamma + F_{ij,t} + \varepsilon_{ij,t},$$

where $\text{PTA}_{ij,t}$ equals one if the pair signs an agreement in year t, and zero otherwise.

We report two benchmark specifications. The first includes country and year fixed effects:

$$F_{ij,t} = \alpha_i + \alpha_j + \lambda_t.$$

The second one includes country-by-year fixed effects:

$$F_{ij,t} = \alpha_{i,t} + \alpha_{j,t}.$$

These absorb country-specific changes in agreement formation while retaining comparisons across trading partners. The control vector contains log distance, contiguity, a common official or primary language, historical colonial ties, log combined GDP, the absolute difference in log GDP, log bilateral trade and log the number of traded HS6 products. Standard errors are clustered by country pair.

The theory predicts $\beta_W < 0$ and $\beta_B > 0$. We also report

$$\delta \equiv \beta_B - \beta_W.$$

δ describes the association when the variation is shifted from within to between sectors, while holding total directional variation fixed. Since $\bar{H} = \bar{W} + \bar{B}$,

$$\beta_W \bar{W} + \beta_B \bar{B} = \delta \bar{B} + \beta_W \bar{H}.$$

Consequently, δ is also the coefficient on between-sector variation in an equivalent speci-

fication controlling for $\bar{H}$. δ helps us answer the earlier question of whether it matters where the variation occurs, conditional on the total variation. It is a difference between estimated coefficients, not an additional regressor.

6.3 Results

Table 1 reports the estimates. The within-sector measure is negatively associated with agreement formation, while between-sector variation is positively associated with it.

Table 1: Bilateral Trade Structure and PTA Formation

	(1) Country and year effects	(2) Country-by-year effects
Within-sector variation, $\bar{W}$	-0.0675^{**}	-0.0377
	(0.0271)	(0.0237)
Between-sector variation, $\bar{B}$	0.0332^{***}	0.0283^{***}
	(0.0101)	(0.0088)
Coefficient difference, $\beta_B - \beta_W$	0.1007^{***}	0.0659^{***}
	(0.0290)	(0.0254)
Geographic, size and bilateral-trade controls	Yes	Yes
Observations	34,830	34,830
Country pairs	5,587	5,587
Signature observations	521	521

Notes: The dependent variable is an indicator for the pair's first PTA signature. Trade measures use bilateral value weights, fractional HS6 trade direction and HS2 sectors, averaged over $t-7$ through $t-5$. Coefficients are percentage-point changes in annual signing probability. Pair-clustered standard errors are in parentheses. The standard error of the coefficient difference accounts for the covariance between the two estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates in column (1) imply that a one-standard-deviation increase in within-sector variation, holding between-sector variation and the controls fixed, is associated with a 0.254 percentage-point lower annual signing probability. This corresponds to approximately 17.0 percent of the sample's average annual signing probability of 1.49 percent. Similarly, a one-standard-deviation increase in between-sector variation, holding within-sector variation and

the controls fixed, is associated with a 0.241 percentage-point higher annual signing probability, equivalent to 16.1 percent of the sample mean. Comparing pairs with the same total directional variation, a one-standard-deviation increase in $B - W$—towards more between-sector and less within-sector variation—is associated with a 0.389 percentage-point higher annual signing probability, equivalent to 26.0 percent of the sample mean. Estimates in column (2) are slightly attenuated, with the magnitudes corresponding, respectively, to 9.5, 13.7 and 17.0 percent of the sample mean.

The two specifications therefore provide consistent evidence on the relative composition margin: conditional on overall directional variation, pairs whose prior trade is more differentiated across sectors are more likely to sign. Evidence for a separate negative within-sector association is less precise with country-by-year effects.

6.4 Robustness

In Appendix F we provide a number of robustness checks. We first show that the results are not sensitive to political economy considerations. We add controls proxying the system of checks and balances, election proximity, and government orientation, which may affect governments' ability and incentives to conclude an agreement.

We also examine sensitivity to timing, product coverage, trade intensity, and agreement type. Earlier and later measurement windows allow for uncertainty about when negotiations begin. Five-year signing horizons reduce reliance on the precise year of signature and the classification of future signers. We further exclude fuels, impose minimum bilateral trade thresholds, and restrict the analysis to first-base-treaty cases. These exercises assess whether the findings are driven by a particular sector, very small trade flows, or the inclusion of extensions and accessions. Taken together, the results broadly preserve the benchmark pattern.

7 Conclusion

The gains from trade liberalisation do not by themselves ensure that countries can agree on how to realise them. The mandates governments fix before talks are themselves chosen strategically, and we have shown that this choice can prevent an efficient agreement even under favourable conditions: with surplus freely transferable and no political, contractual, or enforcement friction, mandates chosen in each side's interest can place every efficient agreement out of reach. We show that whether they do can be organised around a single, second-order property of the environment, the modularity of aggregate welfare. Strict submodularity makes the two governments' unilateral guarantees incompatible with efficiency. Strict supermodularity, together with additional regularity conditions, leaves room for an efficient agreement.

In a trading economy, this condition becomes observable. In the general equilibrium model we provide it is governed by where a country pair's specialisation lies in the product hierarchy. Two-way trade within narrow sectors pushes welfare toward submodularity and works against agreement, while specialisation across sectors pushes it the other way. This distinction has an empirical counterpart: we measure whether bilateral specialisation occurs primarily within or across sectors and relate it to subsequent agreement formation. The illustrative evidence is consistent with the mechanism. Holding a pair's overall balance of trade fixed, pairs specialised across rather than within sectors are more likely to sign.

The mechanism, finally, is not unique to the tariff-setting environment. What the characterisation identifies is a property of underlying strategic environments in which parties commit to contingent transfers before acting. That is, whether transferable surplus suffices for efficiency depends not on how much surplus there is, but on how the parties' actions combine to produce it. We close the paper by showing that the same modularity distinction classifies familiar environments well beyond tariff-setting. In this sense, our results provide

a primitive way to evaluate when the endogenous mechanism-design framework of Jackson and Wilkie (2005) can sustain efficient outcomes (and indeed any outcome): rather than solving the modified-solo-payoff problem case by case, one can in our class of environments look directly to the modularity of aggregate payoffs.

7.1 Beyond Trade Agreements

Cournot Competition. Consider two firms choosing quantities q_i under linear inverse demand $P(Q) = a - bQ$, where $Q = q_1 + q_2$, and constant and symmetric marginal cost c. Firm i's payoff is

$$G_i(q_i, q_j) = \big(a - b(q_i + q_j) - c\big)q_i.$$

An increase in the rival firm's quantity lowers firm i's payoff, while the non-cooperative equilibrium features excessive production relative to the joint-profit-maximising outcome. Aggregate profits satisfy

$$\frac{\partial^2 G}{\partial q_1 \partial q_2} = -2b < 0.$$

Hence, aggregate payoffs are strictly submodular and our impossibility result applies:[14] independently chosen contingent transfers cannot sustain the joint-profit-maximising reduction in output. This reproduces the Cournot example of Jackson and Wilkie (2005) as a special case of the general modularity result.

[14]The impossibility result requires only strict submodularity and the existence of a pure Nash equilibrium whose actions are excessive relative to the efficient profile (which is true in this setting after relabelling); joint concavity, the negative-externality assumption, and the individual-payoff conditions play no role.

Differentiated Bertrand Competition. The opposite case arises naturally under price competition.[15] Suppose firms face linear differentiated demands

$$D_i(p_i, p_j) = a_i - bp_i + dp_j, \qquad b > d > 0,$$

with profits $G_i(p_i, p_j) = (p_i - c_i)D_i(p_i, p_j)$. Aggregate profits satisfy

$$\frac{\partial^2 G}{\partial p_1 \partial p_2} = 2d > 0.$$

Prices generate a positive externality, and non-cooperative competition leads firms to set prices below the joint-profit-maximising level. Relabelling actions as price concessions, $x_i = \bar{p}_i - p_i$, restores the negative-externality and excessive-action formulation of the general model while preserving the sign of the aggregate cross-partial. Aggregate payoffs are therefore strictly supermodular, and, subject to the regularity conditions above, the efficient outcome can be sustained through independently chosen contingent transfers.[16]

Partnerships and Team Production. Consider a standard partnership in which two agents choose costly efforts a_i and jointly produce output $F(a_1, a_2)$. Agent i receives a share λ_i of output, where $\lambda_1 + \lambda_2 = 1$, so that

$$G_i(a_i, a_j) = \lambda_i F(a_1, a_2) - C_i(a_i).$$

[15] Note that Jackson and Wilkie (2005) show that it is possible to divide the efficient surplus in a Bertrand example without differentiated goods.

[16] The full hypotheses of the possibility result hold, in the relabelled actions $x_i = \bar{p}_i - p_i$. The externality is strict, since $\partial G_i/\partial p_j = d\,(p_i - c_i) > 0$ becomes a strict negative externality after relabelling; equilibrium prices lie strictly below the joint-profit-maximising level, so actions are excessive; the aggregate Hessian $\left(\begin{smallmatrix} -2b & 2d \\ 2d & -2b \end{smallmatrix}\right)$ is negative definite because $b > d$, so G is strictly jointly concave; each G_j is linear in the rival's price, so diminishing harm holds with equality; and $\partial^2 G_i/\partial p_1 \partial p_2 = d$ is a non-zero constant, so individual modularity is stable.

Because each agent receives only part of the return generated by their effort, the non-cooperative equilibrium generally features under-provision of effort relative to the joint-surplus-maximising outcome. Suppose efforts are complementary, so that

$$\frac{\partial^2 F}{\partial a_1 \partial a_2} > 0.$$

Aggregate surplus is

$$G(a_1, a_2) = F(a_1, a_2) - C_1(a_1) - C_2(a_2),$$

and therefore

$$\frac{\partial^2 G}{\partial a_1 \partial a_2} = \frac{\partial^2 F}{\partial a_1 \partial a_2} > 0.$$

Relabelling actions as effort shortfalls, $x_i = \bar{a}_i - a_i$, places the game in the negative-externality and excessive-action form of our framework without changing the sign of the aggregate cross-partial. Hence, again subject to the regularity conditions of our paper, efficient effort can be sustained through independently chosen contingent transfers.[17]

This last example also illustrates the tractability of the modularity characterisation. For general nonlinear production and cost functions, applying the Jackson–Wilkie condition requires solving the agents' nested utility guarantee problems. Here, by contrast, the relevant classification follows directly from the primitive technological condition $F_{12} > 0$.

Taken together, the examples show that the distinction identified above is not particular to trade. Quantity competition, price competition, and team production generate different conclusions because they generate different interactions in aggregate surplus. Modularity

[17] Beyond $F_{12} > 0$, the possibility result's hypotheses hold in the relabelled actions $x_i = \bar{a}_i - a_i$ under the conditions set out below. The externality is strict when F is increasing in each effort ($\partial G_i / \partial a_j = \lambda_i F_{a_j} > 0$, a strict negative externality after relabelling); equilibrium effort lies below the surplus-maximising level, so actions are excessive; and $\partial^2 G_i / \partial a_1 \partial a_2 = \lambda_i F_{12}$ has constant sign, so individual modularity is stable. Diminishing harm, $\partial^2 G_j / \partial a_i^2 = \lambda_j F_{a_i a_i} \geq 0$, requires each agent's marginal product to be non-decreasing in own effort; it holds with equality for the benchmark $F(a_1, a_2) = a_1 a_2$, for which G is strictly jointly concave whenever the cost functions satisfy $C_1''' C_2''' > 1$.

therefore provides a simple way of classifying when independently chosen contingent transfers can, and cannot, support efficient cooperation.

Appendix

A Proofs for Section 4

A.1 Proof of Lemma 1

Proof. Fix a target $t' = (t'_i, t'_j) \in T^2_\Delta$ and $\varepsilon > 0$. Consider the deviation to a schedule of the form

$$S'_i(t_i, t_j) \;=\; \underbrace{S_j^{\hat{t},u}(t_i, t_j)}_{\text{(a) return the partner's concessions}} \;+\; \underbrace{q(t_j)}_{\text{(b) purchase the partner's consent}} \;+\; \underbrace{M\mathbf{1}\{t_i \neq t'_i\}}_{\text{(c) commit to own tariff}},$$

where

$$q(t_j) \;:=\; \max_{t''_j \in T_\Delta} G_j(t'_i; t''_j) - G_j(t'_i; t_j) + \varepsilon \mathbf{1}\{t_j = t'_j\}, \qquad M \;>\; \max_{t\in[0,\bar{t}]^2} G_i(t) - \min_{t\in[0,\bar{t}]^2} G_i(t).$$

Each component is non-negative: (a) and (c) by inspection, and (b) because $\max_{t''_j} G_j(t'_i; t''_j) \geq G_j(t'_i; t_j)$ for every t_j.

Step 1: t'_i is strictly dominant for country i. Component (a) exactly cancels the partner's schedule, so country i's induced payoff is

$$G_i(t_i; t_j) + S_j^{\hat{t},u}(t_i, t_j) - S'_i(t_i, t_j) \;=\; G_i(t_i; t_j) - q(t_j) - M\mathbf{1}\{t_i \neq t'_i\}.$$

Notice that $q(t_j)$ does not depend on t_i. It therefore cancels when country i compares its own tariffs at a fixed t_j, and for every $t_i \neq t'_i$ and every t_j,

$$\big[G_i(t'_i; t_j) - q(t_j)\big] - \big[G_i(t_i; t_j) - q(t_j) - M\big] \;=\; G_i(t'_i; t_j) - G_i(t_i; t_j) + M \;>\; 0$$

by the choice of M. Hence t'_i strictly dominates every other tariff, and every (potentially mixed) Nash equilibrium of the induced game places probability one on $t_i = t'_i$.

It is worth noting why component (b) must be paid at every own tariff rather than only at $t_i = t'_i$. Were the consent payment confined to the target row it would depress country i's payoff precisely at t'_i, and taking M above the range of G_i would no longer establish dominance. It is also worth noting why strict dominance is needed rather than strictness of the target as an equilibrium: strictness at t' does not preclude other equilibria elsewhere, and the modified solo payoff would then be evaluated at the worst of them.

Step 2: t'_j is the unique best reply. Given $t_i = t'_i$, country j receives

$$G_j(t'_i; t_j) + S'_i(t'_i, t_j) - S_j^{\hat{t},u}(t'_i, t_j) \;=\; G_j(t'_i; t_j) + q(t_j) \;=\; \max_{t''_j \in T_\Delta} G_j(t'_i; t''_j) + \varepsilon \mathbf{1}\{t_j = t'_j\}.$$

Component (b) has therefore flattened country j's payoff to a constant across all of its tariffs, and the ε tilts it uniquely towards t'_j. Hence t'_j is the unique best reply.

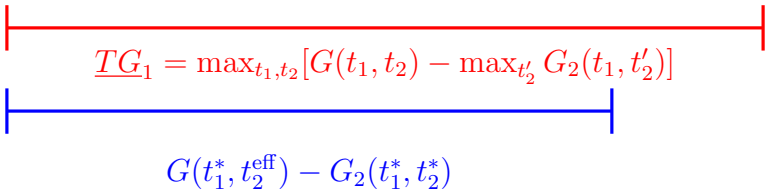
Step 3: the continuation is unique and the value is as claimed. Combining Steps 1 and 2, the induced game has the unique equilibrium t', so the least favourable and the most favourable continuations coincide and the minimum in u_i^{ms} is attained at t'. Country i's payoff there is

$$G_i(t') - q(t'_j) \;=\; G_i(t') - \max_{t''_j} G_j(t'_i; t''_j) + G_j(t') - \varepsilon \;=\; G(t') - \max_{t''_j} G_j(t'_i; t''_j) - \varepsilon.$$

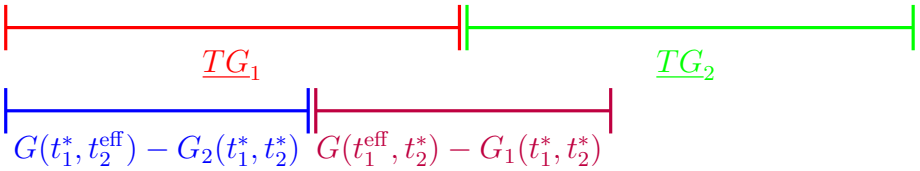
Maximising over targets t' gives $u_i^{ms} \geq \underline{TG}_i - \varepsilon$. Since $\varepsilon > 0$ was arbitrary while u_i^{ms} does not depend on it, the claim follows. □

A.2 Proof of Proposition 1

Proof. $\underline{TG}_i = \max_{t_i,t_j}[G(t_i,t_j) - \max_{t'_j} G_j(t_i,t'_j)]$. As $t^*_j \in \operatorname{argmax}_{t'_j} G_j(t^*_i,t'_j)$, we know that $G(t^*_i,t^{\text{eff}}_j) - G_j(t^*_i,t^*_j) = G(t_i,t_j) - \max_{t'_j} G_j(t_i,t'_j)$ for the pair $(t_i,t_j) = (t^*_i,t^{\text{eff}}_j)$, where t^*_i,t^*_j are Nash equilibrium tariffs. Therefore $\underline{TG}_i \geq G(t^*_i,t^{\text{eff}}_j) - G_j(t^*)$.



Recall that $t^{\text{eff}}_i < t^*_i$ for $i = 1,2$. Therefore $\underline{TG}_1 + \underline{TG}_2 \geq G(t^{\text{eff}}_1,t^*_2) + G(t^*_1,t^{\text{eff}}_2) - G_1(t^*) - G_2(t^*) = G(t^{\text{eff}}_1,t^*_2) + G(t^*_1,t^{\text{eff}}_2) - G(t^*)$.



By strict submodularity we have that $G(t^{\text{eff}}) - G(t^{\text{eff}}_1,t^*_2) < G(t^*_1,t^{\text{eff}}_2) - G(t^*_1,t^*_2)$. Therefore we conclude that $G(t^{\text{eff}}_1,t^*_2) + G(t^*_1,t^{\text{eff}}_2) - G(t^*) > G(t^{\text{eff}})$. With this, $\underline{TG}_1 + \underline{TG}_2 > G(t^{\text{eff}})$. This comparison is the one illustrated in Figure 1 in the main text.

Proving the statement. □

B Proofs for the Positive Result

We first record the construction underlying the possibility result. Fix a division $u_1 + u_2 = G(t^{eff})$. Country j's minimal supporting schedule caps its payoff at u_j on the target row; let

$$\mathcal{R}_j(u_j) \;:=\; \{t_j \in T_\Delta \;:\; G_j(t^{eff}_i;t_j) \geq u_j\}$$

be the tariffs at which that cap binds, country j's *cap-compatible* responses. For a deviator tariff $t_i \in [0, \bar{t}]$, the *modified pure solo claim* is

$$\widehat{TG}_i(t_i \mid u) \;:=\; \max_{t_j \in [0,\bar{t}]} G(t_i; t_j) \;-\; \max_{t_j \in \mathcal{R}_j(u_j)} G_j(t_i; t_j), \qquad \widehat{TG}_i(u) \;:=\; \max_{t_i \in [0,\bar{t}]} \widehat{TG}_i(t_i \mid u).$$

The two maxima run over different sets, and the distinction is essential. The aggregate term maximises over the whole interval $[0, \bar{t}]$, which keeps $\max_{t_j \in [0,\bar{t}]} G(t_i; t_j)$ concave in t_i and so open to Jensen's inequality against a mixed own-tariff: partial maximisation preserves concavity over the convex interval, but not over the grid, where the upper envelope of finitely many concave functions need not be concave. The partner term maximises only over $\mathcal{R}_j(u_j) \subseteq T_\Delta$, crediting the partner solely with grid responses under which its own mandate already secures it u_j, a feasible response in the finite game, which keeps the claim valid when a deviation induces mixed continuation play. The purely continuous counterparts $\underline{TG}_i^c$ and $\underline{TG}_i^c(t_i \mid u)$, defined below, take *both* maxima over the whole interval $[0, \bar{t}]$.

Lemma 2 (Mixed-deviation bound)**.** *Let u satisfy $u_1 + u_2 = G(t^{eff})$. If $\widehat{TG}_i(u) \le u_i$ for $i = 1, 2$, then (t^{eff}, u) is supportable.*

Proof. We verify the Jackson–Wilkie criterion at (t^{eff}, u): with the partner j holding its minimal supporting schedule $S_j^{t^{eff},u}$, no deviation by i yields more than u_i in any continuation. Fix a deviation S_i', let $\mu = \mu_i \times \mu_j$ be a Nash equilibrium of the induced final stage, and write

$$U_i \;:=\; U_i\big(\mu, (S_i', S_j^{t^{eff},u})\big) \;=\; \mathbb{E}_\mu\big[\, G_i(t) + S_j^{t^{eff},u}(t) - S_i'(t) \,\big]$$

for country i's expected payoff. We show $U_i \le u_i$; as S_i' and μ are arbitrary, this gives $u_i^{ms}(t^{eff}, u) \le u_i$, and hence supportability.

Recall that j's schedule pays only when i plays the target t_i^{eff}, and then only j's surplus

above u_j:

$$S_j^{t^{eff},u}(t) \;=\; \max\{G_j(t) - u_j,\, 0\} \cdot \mathbf{1}\{t_i = t_i^{eff}\}.$$

Two consequences are used repeatedly. *(i)* Transfers cancel in the sum of the two governments' payoffs, $TG_i(t,S) + TG_j(t,S) = G(t)$ at every profile. *(ii)* When i plays t_i^{eff}, country j can secure u_j: responding with any cap-compatible $t_j \in \mathcal{R}_j(u_j)$ leaves it with $G_j(t_i^{eff}; t_j) - \big(G_j(t_i^{eff}; t_j) - u_j\big) + S_i'(t) = u_j + S_i'(t) \geq u_j$. Such a response exists because $\max_{t_j} G_j(t_i^{eff}; t_j) \geq u_j$; were it not so, $\mathcal{R}_j(u_j)$ would be empty and $\widehat{TG}_i(t_i^{eff} \mid u)$ ill-defined, which the hypothesis $\widehat{TG}_i(u) \leq u_i$ excludes.

Let $p := \mu_i(t_i^{eff})$ be the probability the deviator's own tariff lands on the target, and let $\bar{t}_i := \mathbb{E}_{\mu_i}[t_i \mid t_i \neq t_i^{eff}]$ denote the mean of its off-target draw. The argument is in *unconditional* expectations: because a mixed continuation commits j to a strategy before i's draw is realised, we do not let j respond to a realised t_i. By *(i)*, $U_i = \mathbb{E}_\mu[G(t)] - U_j$, so it suffices to bound $\mathbb{E}_\mu[G(t)]$ from above and U_j from below.

Aggregate payoff. Splitting at the atom,

$$\mathbb{E}_\mu[G(t)] = p\,\mathbb{E}_{\mu_j}\big[G(t_i^{eff}; t_j)\big] + (1-p)\,\mathbb{E}_\mu\big[G(t_i; t_j) \mid t_i \neq t_i^{eff}\big] \;\leq\; p\,G(t^{eff}) + (1-p) \max_{t_j \in [0,\bar{t}]} G(\bar{t}_i; t_j).$$

The atom term uses $\mathbb{E}_{\mu_j}[G(t_i^{eff}; t_j)] \leq \max_{t_j} G(t_i^{eff}; t_j) = G(t^{eff})$, since t_j^{eff} is the aggregate best reply to t_i^{eff}. The off-atom term uses $G(t_i; t_j) \leq \max_{t_j \in [0,\bar{t}]} G(t_i; t_j)$ pointwise, valid at every grid profile since the interval maximum dominates the grid, together with Jensen's inequality: $\max_{t_j \in [0,\bar{t}]} G(t_i; t_j)$ is concave in t_i, because partial maximisation over the *convex* interval $[0, \bar{t}]$ preserves the joint concavity of G.

Partner's payoff. Fix any single cap-compatible response $\hat{t}_j \in \mathcal{R}_j(u_j)$ and consider j holding it against i's mixed own-tariff. Country j's own schedule pays i only on the atom, and there exactly the surplus $G_j(t_i^{eff}; \hat{t}_j) - u_j \geq 0$ (non-negative as $\hat{t}_j \in \mathcal{R}_j(u_j)$), while i's

schedule pays j a non-negative amount; so this pure response secures j at least

$$\mathbb{E}_{\mu_i}\big[G_j(t_i;\hat{t}_j)\big] - p\big(G_j(t_i^{eff};\hat{t}_j) - u_j\big) \;=\; p\,u_j + (1-p)\,\mathbb{E}_{\mu_i}\big[G_j(t_i;\hat{t}_j) \mid t_i \neq t_i^{eff}\big].$$

By Nash optimality U_j is at least this for every $\hat{t}_j \in \mathcal{R}_j(u_j)$. Diminishing harm makes $G_j(t_i;\hat{t}_j)$ convex in t_i, so Jensen gives $\mathbb{E}_{\mu_i}[G_j(t_i;\hat{t}_j) \mid t_i \neq t_i^{eff}] \geq G_j(\bar{t}_i;\hat{t}_j)$, and taking the best cap-compatible response,

$$U_j \;\geq\; p\,u_j + (1-p) \max_{\hat{t}_j \in \mathcal{R}_j(u_j)} G_j(\bar{t}_i;\hat{t}_j).$$

Combining. Subtracting the two bounds,

$$\begin{aligned} U_i = \mathbb{E}_\mu[G(t)] - U_j \;&\leq\; p\big(G(t^{eff}) - u_j\big) + (1-p)\Big[\max_{t_j \in [0,\bar{t}]} G(\bar{t}_i;t_j) - \max_{\hat{t}_j \in \mathcal{R}_j(u_j)} G_j(\bar{t}_i;\hat{t}_j)\Big] \\ &= p\,u_i + (1-p)\,\widehat{TG}_i(\bar{t}_i \mid u), \end{aligned}$$

using $G(t^{eff}) - u_j = u_i$ and the definition of $\widehat{TG}_i(\bar{t}_i \mid u)$. Since $\widehat{TG}_i(\bar{t}_i \mid u) \leq \max_{t_i} \widehat{TG}_i(t_i \mid u) = \widehat{TG}_i(u) \leq u_i$ by hypothesis, U_i is a convex combination of quantities each at most u_i, so $U_i \leq u_i$. As S_i' and the continuation μ were arbitrary, (t^{eff}, u) is supportable. □

B.1 The Continuous Auxiliary Problem

For the proof it is convenient to work with the continuum counterpart of the pure solo claim, denoted $\underline{TG}_i^c$: the same object as $\underline{TG}_i$, but with the partner's responses ranging over the whole interval $[0,\bar{t}]$ rather than the grid T_Δ. Explicitly, for a deviator tariff $t_i \in [0,\bar{t}]$,

$$\underline{TG}_i^c(t_i) \;:=\; \max_{t_j \in [0,\bar{t}]} G(t_i;t_j) - \max_{t_j \in [0,\bar{t}]} G_j(t_i;t_j), \qquad \underline{TG}_i^c \;:=\; \max_{t_i \in [0,\bar{t}]} \underline{TG}_i^c(t_i),$$

and, for a division u, with $\mathcal{R}_j^c(u_j) := \{t_j \in [0,\bar{t}] : G_j(t_i^{eff}; t_j) \geq u_j\}$,

$$\underline{TG}_i^c(t_i \mid u) \;:=\; \max_{t_j \in [0,\bar{t}]} G(t_i; t_j) - \max_{t_j \in \mathcal{R}_j^c(u_j)} G_j(t_i; t_j),$$

the continuous counterpart of the modified pure solo claim.

We first record two facts that follow from the maintained assumptions and are used repeatedly.

Lemma 3. *Under assumptions 3 and 5 and diminishing harm, for each t_i, $G_j(t_i; t_j)$ is strictly concave in t_j, country j has a unique best reply $b_j(t_i)$, and b_j is continuous and monotone.*

Proof. Write $G_j(t_i; t_j) = G(t_i; t_j) - G_i(t_i; t_j)$. Joint concavity makes $G(t_i; t_j)$ strictly concave in t_j, and diminishing harm applied to country i makes $G_i(t_i; t_j)$ weakly convex in t_j, since t_j is the *other* country's tariff from i's point of view. The difference is therefore strictly concave, so the best reply is unique, and it is continuous by the maximum theorem. Assumption 5 gives monotonicity: b_j is weakly increasing if G_j is supermodular and weakly decreasing if G_j is submodular. □

B.2 The First-Exit Lemma

The following result is the heart of the argument. It says that confining the partner to its cap-compatible responses does not raise the deviator's claim at all, provided the division leaves each country strictly above its solo claim.

Lemma 4 (First exit)**.** *Suppose assumptions 3 and 5 hold, together with diminishing harm, and let u satisfy $u_1 + u_2 = G(t^{eff})$. If $u_i > \underline{TG}_i^c$ for $i = 1, 2$, then $\max_{t_i} \underline{TG}_i^c(t_i \mid u) = \underline{TG}_i^c$ for $i = 1, 2$.*

Proof. Fix country i and let j be the other country.

Step 1: preliminaries. Recall that $\underline{TG}_i^c(t_i \mid u)$ differs from $\underline{TG}_i^c(t_i)$ only in confining the partner's response to $\mathcal{R}_j^c(u_j) = \{t_j : G_j(t_i^{eff}; t_j) \geq u_j\}$. Joint concavity of G implies that $\max_{t_j} G(t_i; t_j)$ is concave in t_i, since partial maximisation preserves concavity. By Lemma 3, $G_j(t_i; t_j)$ is strictly concave in t_j, with a unique, continuous and monotone best reply b_j. Since $G_j(t_i^{eff}; t_j)$ is concave in t_j, the set $\mathcal{R}_j^c(u_j)$ is an interval, which we write as $[\ell_j, r_j]$.

Step 2: the partner's best reply to t_i^{eff} lies in the interior. Evaluating $\underline{TG}_i^c$ at $t_i = t_i^{eff}$ gives $\underline{TG}_i^c \geq \max_{t_j} G(t_i^{eff}; t_j) - \max_{t_j} G_j(t_i^{eff}; t_j) = G(t^{eff}) - \max_{t_j} G_j(t_i^{eff}; t_j)$, since $\max_{t_j} G(t_i^{eff}; t_j) = G(t^{eff})$. Combining with $u_i > \underline{TG}_i^c$ and $u_j = G(t^{eff}) - u_i$,

$$u_j \;<\; \max_{t_j} G_j(t_i^{eff}; t_j) \;=\; G_j\big(t_i^{eff}; b_j(t_i^{eff})\big),$$

so $b_j(t_i^{eff})$ lies in the interior of $\mathcal{R}_j^c(u_j)$.

Step 3: before the first exit. Whenever $b_j(t_i) \in \mathcal{R}_j^c(u_j)$, the constrained and unconstrained maxima coincide, and therefore

$$\underline{TG}_i^c(t_i \mid u) \;=\; \underline{TG}_i^c(t_i) \;\leq\; \underline{TG}_i^c.$$

If this holds for every t_i then the claim follows immediately, so suppose it does not.

Step 4: the first exit. Consider t_i rising above t_i^{eff}, and suppose the best reply eventually leaves $\mathcal{R}_j^c(u_j)$. Since b_j is continuous and monotone and $b_j(t_i^{eff})$ is interior, there is a first exit $t_i^\dagger > t_i^{eff}$, at which $b_j(t_i^\dagger)$ equals the endpoint of $[\ell_j, r_j]$ towards which b_j is moving. That endpoint is r_j when b_j is increasing and ℓ_j when b_j is decreasing; we denote it by t_j^∂ in either case. By definition of the endpoint,

$$G_j(t_i^{eff}; t_j^\partial) \;=\; u_j.$$

At the first exit, t_j^∂ is still an unrestricted best reply, so

$$\max_{t_j} G(t_i^\dagger; t_j) - G_j(t_i^\dagger; t_j^\partial) \;=\; \underline{TG}_i^c(t_i^\dagger) \;\leq\; \underline{TG}_i^c.$$

Step 5: beyond the first exit. For $t_i > t_i^\dagger$, monotonicity carries $b_j(t_i)$ strictly beyond t_j^∂ in the same direction, and t_j^∂ is the endpoint of $\mathcal{R}_j^c(u_j)$ nearest to $b_j(t_i)$. Strict concavity of $G_j(t_i; t_j)$ in t_j therefore makes it monotone between t_j^∂ and $b_j(t_i)$, so the constrained maximum over $\mathcal{R}_j^c(u_j)$ continues to be attained at t_j^∂. Hence

$$\underline{TG}_i^c(t_i \mid u) \;=\; \phi(t_i) \;:=\; \max_{t_j} G(t_i; t_j) - G_j(t_i; t_j^\partial) \qquad \text{for all } t_i \geq t_i^\dagger.$$

Note that ϕ is concave, since $\max_{t_j} G(t_i; t_j)$ is concave in t_i and diminishing harm makes $G_j(t_i; t_j^\partial)$ convex in t_i.

Step 6: the comparison. Evaluate ϕ at the efficient own tariff. Because t^{eff} maximises G jointly, t_j^{eff} is the aggregate best reply to t_i^{eff}, so $\max_{t_j} G(t_i^{eff}; t_j) = G(t^{eff})$; and the endpoint identity of Step 4 gives $G_j(t_i^{eff}; t_j^\partial) = u_j$. Hence

$$\phi(t_i^{eff}) \;=\; \max_{t_j} G(t_i^{eff}; t_j) - G_j(t_i^{eff}; t_j^\partial) \;=\; G(t^{eff}) - u_j \;=\; u_i,$$

the last step using $u_i + u_j = G(t^{eff})$. At the first exit, Step 4 gives $\phi(t_i^\dagger) \leq \underline{TG}_i^c < u_i = \phi(t_i^{eff})$. So ϕ is strictly lower at $t_i^\dagger$ than at t_i^{eff}, and concavity therefore implies that ϕ is non-increasing on $[t_i^\dagger, \bar{t}]$. Hence

$$\phi(t_i) \;\leq\; \phi(t_i^\dagger) \;\leq\; \underline{TG}_i^c \qquad \text{for all } t_i \geq t_i^\dagger.$$

Step 7: the other side, and conclusion. The argument for $t_i < t_i^{eff}$ is identical. Monotonicity again identifies a single endpoint at which the best reply exits, the function ϕ is

again concave with $\phi(t_i^{eff}) = u_i$ and ϕ no greater than $\underline{TG}_i^c$ at the first exit, and concavity again prevents it from rising above $\underline{TG}_i^c$ as t_i moves further from t_i^{eff}. Combining Steps 3, 6 and 7, $\underline{TG}_i^c(t_i \mid u) \leq \underline{TG}_i^c$ for every t_i. Since restricting the partner's response set can only raise the expression, $\max_{t_i} \underline{TG}_i^c(t_i \mid u) \geq \underline{TG}_i^c$, and the two are therefore equal. □

B.3 Strict Slack

Recall $\underline{TG}_i^c$, the continuous pure solo claim defined above, in which the partner's responses range over the whole interval $[0, \bar{t}]$ rather than the grid.

Lemma 5 (Strict slack)**.** *Suppose G is strictly supermodular. Then $\underline{TG}_1^c + \underline{TG}_2^c < G(t^{\text{eff}})$.*

Beyond strict supermodularity, the argument uses the maintained assumptions: the strict negative externality (Assumption 1), joint concavity, and excessive Nash tariffs (Assumption 2).

Proof. Choose (t_1, t_2) and (t_1', t_2') attaining the two claims, so that

$$\underline{TG}_1^c = G(t_1, t_2) - \max_{t_2''} G_2(t_1; t_2''), \qquad \underline{TG}_2^c = G(t_1', t_2') - \max_{t_1''} G_1(t_1''; t_2').$$

Case 1: $t_1' > t_1$. Writing $G(t_1', t_2') = G_1(t_1', t_2') + G_2(t_1', t_2')$ and using the strict negative externality $\max_{t_2''} G_2(t_1; t_2'') > \max_{t_2''} G_2(t_1'; t_2'')$ (as $t_1' > t_1$),

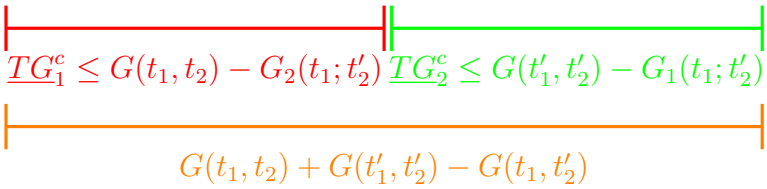
$$\begin{aligned}
\underline{TG}_1^c + \underline{TG}_2^c &= \Big[G(t_1, t_2) - \max_{t_2''} G_2(t_1; t_2'')\Big] + \Big[G(t_1', t_2') - \max_{t_1''} G_1(t_1''; t_2')\Big] \\
&< G(t_1, t_2) + \Big[G_1(t_1', t_2') - \max_{t_1''} G_1(t_1''; t_2')\Big] + \Big[G_2(t_1', t_2') - \max_{t_2''} G_2(t_1'; t_2'')\Big] \\
&\leq G(t_1, t_2) \;\leq\; G(t^{eff}),
\end{aligned}$$

the final line because both bracketed terms are non-positive.

Case 2: $t_2 > t_2'$. The symmetric argument gives $\underline{TG}_1^c + \underline{TG}_2^c < G(t_1', t_2') \leq G(t^{eff})$.

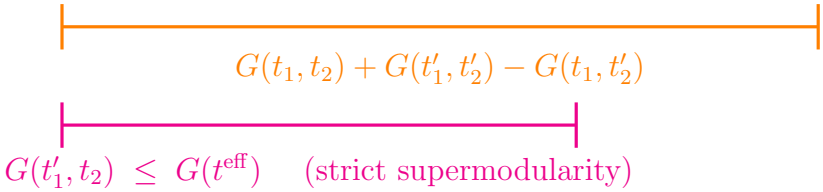
Case 3: $t_1 \geq t_1'$ *and* $t_2' \geq t_2$. This is the main case, and it is the mirror image of the diagrams accompanying Proposition 1: there the two claims overshot $G(t^{eff})$, whereas here they fall short of it. Using $\max_{t_2''} G_2(t_1; t_2'') \geq G_2(t_1; t_2')$ and $\max_{t_1''} G_1(t_1''; t_2') \geq G_1(t_1; t_2')$,

$$\begin{aligned} \underline{TG}_1^c + \underline{TG}_2^c &\leq \big[G(t_1, t_2) - G_2(t_1; t_2')\big] + \big[G(t_1', t_2') - G_1(t_1; t_2')\big] \\ &= G(t_1, t_2) + G(t_1', t_2') - \big[G_1(t_1; t_2') + G_2(t_1; t_2')\big] \\ &= G(t_1, t_2) + G(t_1', t_2') - G(t_1, t_2'). \end{aligned} \tag{1}$$



Suppose first that $t_1 > t_1'$ and $t_2' > t_2$, so that the rectangle spanned by (t_1', t_2) and (t_1, t_2') is non-degenerate. Strict supermodularity gives $G(t_1, t_2) + G(t_1', t_2') < G(t_1, t_2') + G(t_1', t_2)$. Substituting into (1),

$$\begin{aligned} \underline{TG}_1^c + \underline{TG}_2^c &\leq G(t_1, t_2) + G(t_1', t_2') - G(t_1, t_2') \\ &< \big[G(t_1, t_2') + G(t_1', t_2)\big] - G(t_1, t_2') \\ &= G(t_1', t_2) \;\leq\; G(t^{eff}). \end{aligned}$$



Case 4: the degenerate rectangles. Suppose $t_1 = t'_1$ and $t'_2 > t_2$. Then (1) reduces to $\underline{TG}_1^c + \underline{TG}_2^c \leq G(t_1, t_2) \leq G(t^{eff})$, and we show the inequality is strict. If it were not, then $G(t_1, t_2) = G(t^{eff})$, and strict joint concavity would give $(t_1, t_2) = t^{eff}$. Equality would also require equality in the two bounds used for (1), that is, t'_2 maximises $G_2(t_1; t_2)$ over t_2 and t_1 maximises $G_1(t_1; t'_2)$ over t_1, so that (t_1, t'_2) is a Nash equilibrium of the tariff-setting game. But then $t_1 = t_1^{eff}$, so this Nash equilibrium is not excessive in the first coordinate, contradicting Assumption 2. The case $t_1 > t'_1$ and $t'_2 = t_2$ is symmetric. Finally, if $t_1 = t'_1$ and $t_2 = t'_2$, the same equalities would make $(t_1, t_2) = t^{eff}$ itself a Nash equilibrium, again contradicting Assumption 2. □

Note that it is here, and only here, that the full strength of Assumption 2 is used. The impossibility result needs only that the non-cooperative equilibrium t^* itself be excessive. Case 4 above instead requires that no Nash equilibrium coincide with the efficient profile in either coordinate, which is the additional content of Assumption 2.

B.4 Proof of Proposition 2

Proof. By Lemma 5 the slack

$$\delta \;:=\; G(t^{eff}) - \underline{TG}_1^c - \underline{TG}_2^c$$

is strictly positive. Set $u_i = \underline{TG}_i^c + \delta/2$ for $i = 1, 2$, so that $u_1 + u_2 = G(t^{eff})$ and $u_i > \underline{TG}_i^c$ for both countries.

Lemma 4 then gives $\max_{t_i} \underline{TG}_i^c(t_i \mid u) = \underline{TG}_i^c$. That is, at this division, confining the partner to its cap-compatible responses does not raise the continuous claim at all.

It remains to pass from the interval to the grid, and it is here that care is required. Unlike the impossibility result, which compared payoffs at the fixed profiles t^{eff} and t^* and

so held on *any* admissible grid, the possibility result compares the hybrid claim, $\widehat{TG}_i(u) = \max_{t_i} \widehat{TG}_i(t_i \mid u)$, against the continuous claim $\underline{TG}_i^c$, and the two need not coincide on a coarse grid. We show that the strict slack δ is exactly what absorbs the discrepancy.

The hybrid claim $\widehat{TG}_i$ differs from the continuous claim $\underline{TG}_i^c$ in *one* place only: the aggregate maximum (over $t_j \in [0, \bar{t}]$) and the outer maximum (over $t_i \in [0, \bar{t}]$) are common to both, and only the partner's response set differs, ranging over the grid $\mathcal{R}_j(u_j) = \mathcal{R}_j^c(u_j) \cap T_\Delta$ in $\widehat{TG}_i$ and over the interval $\mathcal{R}_j^c(u_j)$ in $\underline{TG}_i^c$. Let ω be a joint modulus of continuity for G and G_j on the compact square $[0, \bar{t}]^2$. By Step 2 of Lemma 4 the interval $\mathcal{R}_j^c(u_j) = [\ell_j, r_j]$ has non-empty interior, so once $m(T_\Delta) < r_j - \ell_j$ every point of $\mathcal{R}_j^c(u_j)$ lies within $m(T_\Delta)$ of a point of $\mathcal{R}_j(u_j)$; in particular $\mathcal{R}_j(u_j)$ is non-empty. Restricting the partner's maximum to this grid subset lowers it by at most $\omega\big(m(T_\Delta)\big)$, and so raises the subtracted term by at most that much, giving the *one-sided* upper bound

$$\widehat{TG}_i(t_i \mid u) \;\le\; \underline{TG}_i^c(t_i \mid u) + \omega\big(m(T_\Delta)\big) \qquad \text{for every } t_i.$$

Taking the maximum over t_i and using $\max_{t_i} \underline{TG}_i^c(t_i \mid u) = \underline{TG}_i^c$ (Lemma 4),

$$\widehat{TG}_i(u) \;\le\; \underline{TG}_i^c + \omega\big(m(T_\Delta)\big).$$

Choose $\bar{\Delta} > 0$ so that $\omega(\bar{\Delta}) < \delta/2$. Then every admissible grid with mesh $m(T_\Delta) < \bar{\Delta}$ satisfies

$$\widehat{TG}_i(u) \;\le\; \underline{TG}_i^c + \omega\big(m(T_\Delta)\big) \;<\; \underline{TG}_i^c + \tfrac{\delta}{2} \;=\; u_i, \qquad i = 1, 2.$$

The discretisation error is thus strictly dominated by the slack that strict supermodularity delivered through Lemma 5. Lemma 2 then implies that (t^{eff}, u) is supportable, and the tariff profile t^{eff} is implemented in a subgame perfect equilibrium. □

C The Modular Boundary

We record the knife-edge case in which aggregate payoff is exactly modular, $\partial^2 G/\partial t_1 \partial t_2 = 0$. Here strict supermodularity is lost, so Lemma 5 no longer delivers strict slack, and the argument of Proposition 2 is unavailable. Supporting t^{eff} then requires the stronger condition that *each* individual payoff be modular.

Proposition 4 (The modular knife edge)**.** *Suppose G is modular and each individual payoff G_i is modular. Then there is a subgame perfect equilibrium implementing t^{eff} on every admissible tariff grid.*

The argument is brief. When each G_i is modular, each country's best reply does not depend on its partner's tariff, so the cap-compatible restriction removes nothing and the grid subtleties of Proposition 2 do not arise. Aggregate modularity, meanwhile, forces the two directions of the earlier arguments to coincide: the submodular construction of Proposition 1 gives $\underline{TG}_1 + \underline{TG}_2 \geq G(t^{eff})$ with equality, while the slack argument of Lemma 5 gives the reverse, so that $\underline{TG}_1 + \underline{TG}_2 = G(t^{eff})$. The pure solo constraints therefore pin a unique division, and individual modularity makes that division supportable.

D The Nested-CES Foundation: Cross Derivative and Assumptions

This appendix derives the cross derivative of aggregate welfare stated in Proposition 3 and records the conditions under which the reduced-form payoffs of Section 5 inherit the maintained assumptions. We present the backbone of the derivation; the full step-by-step algebra is omitted as it is mostly standard.

Throughout write $\Gamma := \gamma_1 + \gamma_2$ and $r_i := \gamma_i/\Gamma$, and, at free trade, define the within- and

across-sub-sector expenditure-share covariances

$$\mathrm{Cov}_k^s(z,y) := \sum_l s_{kl} z_l y_l - \Big(\sum_l s_{kl} z_l\Big)\Big(\sum_l s_{kl} y_l\Big),$$
$$\mathrm{Cov}^\theta(Z,Y) := \sum_k \theta_k Z_k Y_k - \Big(\sum_k \theta_k Z_k\Big)\Big(\sum_k \theta_k Y_k\Big).$$

D.1 Derivation of the Cross Derivative

Recall from Section 5 that aggregate welfare depends on tariffs only through the two price indices, $G = \bar{x}_1 + \bar{x}_2 + \sum_i \gamma_i \log \gamma_i - \sum_i \gamma_i \log P_i$, so that

$$\frac{\partial^2 G}{\partial t_1 \partial t_2} = -\sum_i \gamma_i \frac{\partial^2 \log P_i}{\partial t_1 \partial t_2}.$$

The derivation has four steps: the Hessian of a nested price index; the general-equilibrium reduction that removes the endogenous second-order price response; the first-order incidence of tariffs on world prices; and the collapse to trade-structure moments. We give the milestones and the reasoning that links them; the full algebra is in the companion notes.

Step 1 (price-index Hessian). The domestic price of good (k,l) in country i is $\log p_{i,kl} = m_{i,kl}\, t_i + \log p^w_{kl}$, so its response to tariff t_j is $h_{i,kl,j} := \partial \log p_{i,kl}/\partial t_j = m_{i,kl}\mathbf{1}\{i=j\} + q_{kl,j}$, where $q_{kl,j} := \partial \log p^w_{kl}/\partial t_j$. Two-stage budgeting gives the nested-CES expenditure shares θ_k and s_{kl}, and Shephard's lemma gives $\partial \log P_i/\partial \log p_{i,kl} = \theta_k s_{kl}$. Differentiating $\log P_i$ once more, and using that under CES a share moves with a log price by $(1-\sigma)$ times that price's deviation from its group average, the Hessian sorts by tier:

$$\frac{\partial^2 \log P_i}{\partial t_1 \partial t_2} = \sum_{k,l} \theta_k s_{kl} \kappa_{kl} + \sum_k \theta_k (1-\sigma_k)\, \mathrm{Cov}_k^s(h_{i\cdot,1}, h_{i\cdot,2}) + (1-\sigma_0)\, \mathrm{Cov}^\theta(\rho_{i\cdot,1}, \rho_{i\cdot,2}),$$

where $\rho_{i,k,j} := \sum_l s_{kl} h_{i,kl,j}$ is the sub-sector-average price response and $\kappa_{kl} := \partial^2 \log p_{i,kl}/\partial t_1 \partial t_2$

is the own second-order term.

Step 2 (general-equilibrium reduction). Sum $-\gamma_i$ times the Step-1 expression over $i = 1, 2$. The domestic-price responses split as $h_{i,kl,j} = m_{i,kl}\mathbf{1}\{i = j\} + q_{kl,j}$, so the mechanical own-tariff pieces would remain were the world prices held fixed; but they are endogenous. Differentiating the aggregate value-clearing identity (world expenditure on each good equals the value of its supply) to second order cancels the own-curvature term $\sum \theta_k s_{kl} \kappa_{kl}$ and the direct wedge, leaving only the world-price responses $q_{kl,j}$ and their sub-sector and economy-wide averages $\pi_{k,j} := \sum_l s_{kl} q_{kl,j}$, $\bar{\pi}_j := \sum_k \theta_k \pi_{k,j}$:

$$\frac{\partial^2 G}{\partial t_1 \partial t_2} = \Gamma \Big[\sum_k \sigma_k \theta_k \operatorname{Cov}_k^s(q_{\cdot,1}, q_{\cdot,2}) + \sigma_0 \operatorname{Cov}^\theta(\pi_1, \pi_2) + \bar{\pi}_1 \bar{\pi}_2 \Big].$$

The coefficients are now σ_k and σ_0; the extra term $\bar{\pi}_1 \bar{\pi}_2$ is the value-of-resources effect generated by the equilibrium price movements.

Step 3 (world-price incidence). The responses q solve the first-order market-clearing conditions. Because preferences are identical and differentiated expenditure is fixed at γ_i, at free trade every country spends the same fraction $r_i = \gamma_i / \Gamma$ of income on each good; the demand side of the clearing condition for good (k, l) is then a common function of its own price, the system decouples across goods, and solving it gives

$$q_{kl,j} = -r_j \, m_{jkl}.$$

A tariff by country j lowers the world price of each good it imports by the fraction r_j and leaves its export prices unchanged to first order. Hence $\pi_{k,j} = -r_j \bar{m}_k^j$ and $\bar{\pi}_j = -r_j \mu^j$, with $\bar{m}_k^j$ and μ^j country j's within-sub-sector and aggregate import shares.

Step 4 (closed form). Let $m_{kl} := m_{1kl}$, so $m_{2kl} = 1 - m_{kl}$, and write $\bar{m}_k := \sum_l s_{kl} m_{kl}$ and $\mu := \sum_k \theta_k \bar{m}_k$. Substituting Step 3, each covariance carries a factor $r_1 r_2$, and because m_{kl}

is $\{0,1\}$-valued the within-sub-sector covariance collapses, $\mathrm{Cov}^s_k(m_{\cdot,1}, m_{\cdot,2}) = \mathrm{Cov}^s_k(m, 1-m) = -\bar{m}_k(1-\bar{m}_k)$, and likewise across sub-sectors; the value-of-resources term contributes $r_1 r_2\, \mu(1-\mu)$. Collecting,

$$\frac{\partial^2 G}{\partial t_1 \partial t_2} = \Gamma\, r_1 r_2 \Big[\mu(1-\mu) - \sum_k \sigma_k \theta_k\, \bar{m}_k(1-\bar{m}_k) - \sigma_0\, \mathrm{Var}^\theta(\bar{m}_k)\Big].$$

The nested law of total variance, $\sum_k \theta_k \bar{m}_k(1-\bar{m}_k) + \mathrm{Var}^\theta(\bar{m}_k) = \mu(1-\mu)$, distributes the leading $\mu(1-\mu)$ across the two channels, and $\Gamma r_1 r_2 = \gamma_1\gamma_2/\Gamma$ gives

$$\frac{\partial^2 G}{\partial t_1 \partial t_2} = \frac{\gamma_1\gamma_2}{\Gamma}\Big[\sum_k \theta_k(1-\sigma_k)\bar{m}_k(1-\bar{m}_k) + (1-\sigma_0)\,\mathrm{Var}^\theta(\bar{m}_k)\Big],$$

which is Proposition 3. □

D.2 Conditions and the Maintained Assumptions

The closed form relies on: (i) a freely traded linear outside good with an interior solution; (ii) fixed differentiated-industry expenditure γ_i; (iii) identical nested-CES preferences across countries; (iv) evaluation at free trade; (v) locally fixed import directions, with exactly one importer per differentiated good; and (vi) local regularity and uniqueness of the differentiated-good price equilibrium. If countries have different CES weights or elasticities they do not purchase the same fraction of every good at free trade, and the simple incidence formula of Step 3 need not hold; if differentiated expenditure responds to income, the outside good no longer absorbs all wealth effects and additional terms appear.

Under these conditions the reduced-form payoffs $G_i(t) = Y_i - \gamma_i + \gamma_i \log\gamma_i - \gamma_i \log P_i$ inherit the assumptions of Sections 3 and 4.3 locally around free trade. Two facts are used repeatedly. First, the first-order world-price incidence of Step 3, $q_{kl,j} = \partial \log p^w_{kl}/\partial t_j = -r_j m_{jkl}$ at free trade: a tariff by country j moves only the world prices of the goods it

imports, lowering each by the fraction r_j. Second, because preferences are quasi-linear in the freely traded numeraire the marginal value of income is one, so the envelope theorem gives, for a change in world prices holding country i's own tariff fixed,

$$\mathrm{d}G_i = \sum_{k,l} \left(e_{i,kl} - c_{i,kl}\right) \mathrm{d}p^w_{kl},$$

the consumption and revenue responses entering only at second order.

The verification uses three further local conditions, each a transparent sufficient condition for one of the individual-payoff assumptions: (vii) the differentiated goods are genuinely substitutable at free trade, with finite elasticities and strictly positive trade, so that a marginal tariff strictly distorts consumption; (viii) world-price pass-through does not steepen, $\partial^2 p^w_{kl}/\partial t_i^2 \geq 0$ on the goods each country imports; and (ix) the individual cross-partials $\partial^2 G_i/\partial t_1 \partial t_2$ are non-zero at free trade. These deliver strict joint concavity, diminishing harm, and stable individual modularity respectively.

Proposition 5 (Local verification)**.** *Under conditions (i)–(ix) there is $\bar{t} > 0$ such that, on $[0, \bar{t}]^2$, the payoffs G_1, G_2, G satisfy the negative-externality, excessive-Nash-tariff and joint-concavity assumptions of Section 3, together with the diminishing-harm and stable-individual-modularity assumptions of Section 4.3, and $t^{\text{eff}} = (0, 0)$.*

Proof. Throughout, "at free trade" means $t = 0$, where domestic and world prices coincide and the two countries share the expenditure shares θ_k, s_{kl}; the extension to $[0, \bar{t}]^2$ is by continuity of the relevant derivatives for $\bar{t}$ small.

Efficiency of free trade. Aggregate welfare is $G = \bar{x}_1 + \bar{x}_2 + \sum_i \gamma_i \log \gamma_i - \sum_i \gamma_i \log P_i$, so income and tariff-revenue transfers net out in aggregate and only consumption-price distortions remain. Free trade removes every wedge between domestic and world prices and so maximises G; hence $t^{\text{eff}} = (0, 0)$ and $\nabla G(0) = 0$.

Negative externality. Fix $j \neq i$. A tariff by j moves only the world prices of the goods j imports, and those are precisely the goods i exports ($m_{ikl} = 1 - m_{jkl} = 0$). The envelope identity gives, at free trade,

$$\frac{\partial G_i}{\partial t_j}\Big|_0 = \sum_{k,l} (e_{i,kl} - c_{i,kl})\, p^w_{kl}\, q_{kl,j} = -\, r_j \sum_{(k,l):\, m_{jkl}=1} p^w_{kl}\, (e_{i,kl} - c_{i,kl}) \;<\; 0,$$

since i is a net exporter, $e_{i,kl} > c_{i,kl}$, of every good j imports. The partner's tariff depresses the world price of i's export goods by r_j, and i loses that fraction of the value of its net exports of them. By continuity the inequality persists on $[0, \bar{t}]^2$.

Excessive Nash tariffs. At free trade a marginal own tariff has, by the envelope theorem, only its terms-of-trade effect,

$$\frac{\partial G_i}{\partial t_i}\Big|_0 = r_i \sum_{(k,l):\, m_{ikl}=1} p^w_{kl}\, (c_{i,kl} - e_{i,kl}) \;>\; 0,$$

the value of i's net imports scaled by r_i: the own tariff improves i's terms of trade at first order and distorts consumption only at second order. Country i's best reply to any partner tariff is therefore a strictly positive tariff, interior if the second-order distortion overtakes the terms-of-trade gain within $[0, \bar{t}]$ and the upper bound $\bar{t}$ otherwise, and for $\bar{t}$ small this holds throughout $[0, \bar{t}]$. Hence every Nash equilibrium has $\hat{t}_i > 0 = t_i^{\text{eff}}$ for both countries.

Joint concavity. Free trade is a critical point of G: by the two calculations above, country i's marginal terms-of-trade gain from its own tariff exactly offsets country j's loss, since i's net imports equal j's net exports on each good by market clearing, so $\nabla G(0) = 0$. Because free trade is efficient, $D^2 G(0)$ is negative semi-definite, and this holds for every admissible elasticity configuration: the differentiated-good aggregators are concave whether sub-sectors are substitutes or complements ($\sigma_0 < 1$), so the elasticities govern only the sign of the cross-partial $\partial^2 G / \partial t_1 \partial t_2$, not whether free trade maximises. The Hessian is negative definite

whenever the goods are genuinely substitutable (finite elasticities and positive trade), so that a marginal tariff strictly distorts consumption at second order and no tariff perturbation is distortion-free. Then G is strictly concave near free trade, and strict joint concavity holds on $[0, \bar{t}]^2$ for $\bar{t}$ small.

Diminishing harm. Consider $G_j(t_i; t_j)$ as a function of t_i, whose first derivative is the net-export valuation $\partial G_j/\partial t_i = \sum_{k,l}(e_{j,kl} - c_{j,kl})\,\partial p^w_{kl}/\partial t_i$. Differentiating again, the leading second-order term at free trade is $-\sum_{k,l}(\partial c_{j,kl}/\partial t_i)\,(\partial p^w_{kl}/\partial t_i)$. Country i's tariff lowers the world price of the goods it imports ($\partial p^w_{kl}/\partial t_i < 0$ there), and those are goods j exports and also consumes, whose demand rises as their price falls ($\partial c_{j,kl}/\partial t_i > 0$); every product in the sum is therefore non-negative. The remaining term is $\sum_{k,l}(e_{j,kl} - c_{j,kl})\,\partial^2 p^w_{kl}/\partial t_i^2$: on the goods i imports, j is a net exporter ($e_{j,kl} - c_{j,kl} > 0$), so this term is non-negative under the primitive condition that world-price pass-through does not steepen, $\partial^2 p^w_{kl}/\partial t_i^2 \geq 0$, so the fall in the world price does not accelerate in the tariff. This holds with equality in the linear benchmark and is the natural sufficient condition for diminishing harm. Under it both terms are non-negative, so $\partial^2 G_j/\partial t_i^2 \geq 0$ near free trade, verifying Assumption 4 on $[0, \bar{t}]^2$ for $\bar{t}$ small.

Stable individual modularity. Each individual cross-partial $\partial^2 G_i/\partial t_1 \partial t_2$ is a smooth function of the trade-structure primitives and continuous in t. We impose, as a local condition, that it is non-zero at free trade; a non-zero value then keeps its sign on a neighbourhood, so taking $\bar{t}$ small each G_i is either weakly super- or weakly submodular throughout $[0, \bar{t}]^2$, with a sign that may differ across i. Only the stability of the sign, not its direction, is used. □

The signs of the two individual cross-partials $\partial^2 G_i/\partial t_1 \partial t_2$ are the disaggregated analogues of the aggregate object computed above; only their *stability*, not their direction, enters the results of Section 4.3, and the economically relevant classification is governed by the aggregate cross-partial of Proposition 3.

E Data Sources and Construction

This appendix documents the sources, cleaning rules, and sample restrictions underlying the empirical analysis. The unit of observation is an unordered country pair in an outcome year. We construct trade measures independently of agreement status and subsequently merge them with agreement histories and control variables.

E.1 Bilateral Trade Data

The benchmark uses UN Comtrade merchandise exports accessed through WITS. The archived files cover 2002–2020 in the 2002 Harmonized System classification. We use exporter-reported flows in each direction; we neither substitute mirror import reports nor add the separately downloaded re-export series. WITS reports trade values in thousands of current US dollars, which we convert to dollars.

We retain six-digit product entries in HS chapters 01–97, excluding the reserved chapter 77. Aggregate entries and special or unallocated codes outside this range are excluded. Fuels, in chapter 27, remain in the benchmark.

Bilateral measures require usable detailed export data from both countries in a given year. Conditional on this reporting requirement, an absent product flow in one direction is treated as zero. A missing reporter-year is not treated as zero trade. This is a minimum availability requirement, not a guarantee of complete reporting.

E.2 Agreements

Agreement information comes from DESTA version 2.2. A qualifying event is a substantive agreement with resolved DESTA scope category 2 or 3 and an observed entry-into-force year.

For each pair, we identify its earliest qualifying signature using the full available agree-

ment history, before restricting the data to the trade sample. Consequently, agreements signed before 2002 are not lost when constructing treatment status. The outcome records the first signature of a subsequently effective substantive agreement, not its entry into force or the initiation of negotiations.

E.3 Economic and Political Economy Controls

Geographical and economic-size controls come from CEPII's Gravity database, version V202211. We retain records for which both countries exist in the source year, exclude self-pairs, and verify that the two directed records agree before retaining one unordered pair.

The controls comprise log distance between the countries' most-populated cities, contiguity, a common official or primary language, a historical colonial or dependency relationship, log combined GDP, and the absolute difference in log GDP. GDP is measured in current thousands of US dollars. These variables are constructed annually and averaged over the same $t-7$ to $t-5$ window.

Two further controls are constructed from the cleaned trade data: gross bilateral trade and the number of distinct HS6 products traded in either direction. Each product is counted once, irrespective of whether it is traded in one or both directions. We average trade values and product counts over the three-year window and then take logs. Thus, the GDP controls average annual log transformations, whereas the trade and product-count controls take logs of their annual averages.

Political controls use the project's existing cleaned extract of the Database of Political Institutions, documented as DPI2020. They comprise the mean checks-and-balances index across the two countries, an indicator that either executive has at most one year remaining in the current term, and an indicator that either government is classified as right-wing. The term-length variable proxies election proximity rather than recording an observed election

date. We construct the controls annually and average them over $t-7$ to $t-5$, requiring both countries' inputs in every year.

F Empirical Exercise: Robustness

We examine robustness to political controls, the timing of trade measurement and agreement formation, and the coverage of products, trading relationships, and agreements. Unless stated otherwise, we retain the benchmark trade measures, economic controls, and country and year fixed effects. Standard errors are clustered by country pair throughout.

F.1 Political Economy

Domestic politics may influence both trade structure and the willingness to conclude an agreement. We therefore add controls from the Database of Political Institutions for institutional checks and balances, election proximity, and government orientation. These capture, respectively, constraints on executive discretion, political horizons, and differences in policy priorities. For each pair-year, we construct the average checks-and-balances index across the two countries, an indicator that either executive has at most one year remaining in the current term, and an indicator that either government is classified as right-wing. We average all three measures over $t-7$ to $t-5$, matching the benchmark trade-measure window.

Table 2 reports both fixed-effect specifications on the same sample with complete political data. Country-by-year effects absorb the checks-and-balances measure, which is an average of country-specific characteristics. The election-proximity and government-orientation measures retain bilateral variation because they depend jointly on the two countries' political circumstances.

Within-sector variation is negatively associated with signing in both specifications, while

Table 2: Political Controls and PTA Formation

	(1) Country and year effects	(2) Country-by-year effects
Within-sector variation, $\bar{W}$	-0.1452^{***}	-0.1017^{***}
	(0.0328)	(0.0274)
Between-sector variation, $\bar{B}$	0.0218	0.0209^{*}
	(0.0133)	(0.0112)
Coefficient difference, $\beta_B - \beta_W$	0.1670^{***}	0.1226^{***}
	(0.0341)	(0.0285)
Geographic, size and bilateral-trade controls	Yes	Yes
Checks and balances	Yes	Absorbed
Election proximity and government orientation	Yes	Yes
Observations	24,746	24,746
Country pairs	4,029	4,029
Countries	136	136
Signature observations	467	467
Outcome years	2009–2019	2009–2019
Mean annual signing probability	1.887%	1.887%

Notes: Linear probability regressions of first PTA signature in year t. Both columns retain the benchmark construction of $\bar{W}$ and $\bar{B}$: Comtrade flows, HS2 nests, fractional trade directions, bilateral product weights, and averages over $t-7$ to $t-5$. Political controls are averaged over the same window; both countries must have complete political data in all three years. Standard errors clustered by country pair are reported in parentheses. The coefficient difference is calculated from the jointly estimated coefficients, with its standard error accounting for their covariance. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

between-sector variation has a positive but less precisely estimated coefficient. The difference $\beta_B - \beta_W$ is positive and significant at the one-percent level in both columns, consistent with a greater signing probability when directional variation lies between rather than within sectors, holding its total fixed. Relative to the main results, the within-sector coefficient is larger in magnitude and the between-sector coefficient is smaller. These changes also reflect the restricted sample due to the availability of the political economy data.

F.2 Robustness to timing

We first move the three-year trade-measurement window from the benchmark $t-7:t-5$ to either $t-9:t-7$ or $t-5:t-3$, retaining the annual first-signature outcome (Panel A in Table 3). Trade measures and time-varying controls are constructed over the corresponding window.

We then replace annual signing with an indicator for signing at any point during a five-year period. Pairs must have no prior qualifying agreement at the start of that period. For the 2005–2009 and 2010–2014 horizons, trade structure and controls are measured over 2002–2004 and 2007–2009, respectively. Controls are pairs that do not sign during the relevant horizon. The "no future signers" variants additionally exclude controls with a recorded first signature by 2019.

Finally, we pool the 2005–2009, 2010–2014, and 2015–2019 horizons, using 2012–2014 trade structure for the last period. Eligible pairs can appear in more than one period. These stacked regressions include country and period fixed effects. In their no-future-signers variant, a pair signing by 2019 can contribute its treated-period observation but cannot contribute earlier periods as a control.

The earlier annual window produces a coefficient difference of 0.2023, roughly twice the benchmark estimate. The later window produces a substantially smaller difference, 0.0267,

Table 3: Robustness to timing

Specification	β_W	β_B	$\beta_B - \beta_W$	N
Panel A: Annual first signature				
$t-9:t-7$; available sample	-0.1505^{***}	0.0518^{***}	0.2023^{***}	25,656
	(0.0347)	(0.0142)	(0.0378)	
$t-5:t-3$; available sample	-0.0120	0.0146^{*}	0.0267	44,995
	(0.0235)	(0.0081)	(0.0250)	
Panel B: First signature within five years				
2010–2014	-0.5695^{***}	0.2407^{***}	0.8102^{***}	3,523
	(0.1646)	(0.0594)	(0.1735)	
2010–2014; no future signers	-0.5321^{***}	0.2591^{***}	0.7912^{***}	3,280
	(0.1850)	(0.0622)	(0.1957)	
2005–2009	-0.0817	0.1736^{**}	0.2554	1,623
	(0.1607)	(0.0729)	(0.1815)	
2005–2009; no future signers	-0.0590	0.2072^{**}	0.2662	1,317
	(0.1984)	(0.0852)	(0.2253)	
Stacked horizons	-0.1388	0.1252^{***}	0.2640^{**}	8,853
	(0.0964)	(0.0365)	(0.1036)	
Stacked; no future signers	-0.1188	0.1601^{***}	0.2789^{**}	8,304
	(0.1210)	(0.0422)	(0.1308)	

Notes: All regressions include the benchmark economic controls and country fixed effects. Panel A additionally includes year effects; the stacked specifications include period effects. Panel A counts pair-year observations; Panel B counts pairs within a single horizon or pair-period observations when stacked. Panel B coefficients concern five-year signing probabilities and are not annualised. "No future signers" excludes control observations with a recorded first signature by 2019. Country-pair-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

which is imprecisely estimated. Because these specifications use different available samples, this contrast cannot be attributed to the measurement lag alone.

The five-year estimates are larger in absolute magnitude, but concern a different outcome. For example, a 0.01 reallocation from within-to between-sector variation is associated with a 0.8102 percentage-point higher signing probability during 2010–2014, compared with 0.1007 percentage points annually in the benchmark. These correspond to approximately 9.7 and 6.7 percent of their respective sample mean signing probabilities. The coefficient difference

is positive and significant for 2010–2014 and for the stacked horizons, including when future signers are excluded from controls. It is less precisely estimated for 2005–2009. The pooled result should therefore not be read as evidence of an equally strong association in every period.

F.3 Product coverage, trade intensity, and agreement type

We first exclude fuels, defined as HS chapter 27, and reconstruct the trade measures using weights renormalised over the remaining products. We use the observations available for both the full and non-fuel baskets, while retaining the original full-basket Comtrade controls for bilateral trade and the number of traded products.

We next restrict the sample to relationships with average annual gross bilateral trade of at least US$1 million or US$10 million over $t-7:t-5$. Gross trade sums flows in both directions, and the thresholds are expressed in current US dollars. These restrictions check whether the results depend on very small trading relationships.

Finally, we retain signer pairs only when their first-signature year includes an original base treaty, rather than only accessions or consolidated records.

The coefficient difference remains positive and statistically significant in all four specifications, ranging from 0.0833 to 0.1185, compared with 0.1007 in the benchmark. Excluding fuels slightly strengthens the estimated composition association, while the US$1 million threshold leaves its magnitude almost unchanged. The higher trade threshold and the base-treaty restriction reduce the magnitude moderately. Under the US$10 million threshold, the within-sector coefficient is imprecise, although the between-sector coefficient and their difference remain positive and statistically significant.

Table 4: Product coverage, trade intensity, and agreement type

Specification	β_W	β_B	$\beta_B - \beta_W$	N
Exclude fuels; original CT controls	-0.0796^{***}	0.0389^{***}	0.1185^{***}	34,681
	(0.0273)	(0.0103)	(0.0296)	
Average trade $\geq$ US$1m	-0.0593^{**}	0.0387^{***}	0.0980^{***}	26,938
	(0.0292)	(0.0118)	(0.0314)	
Average trade $\geq$ US$10m	-0.0313	0.0520^{***}	0.0833^{**}	18,956
	(0.0334)	(0.0153)	(0.0366)	
First-base-treaty cases only	-0.0603^{**}	0.0276^{***}	0.0878^{***}	34,547
	(0.0263)	(0.0099)	(0.0282)	

Notes: The dependent variable indicates an annual first qualifying PTA signature. All regressions include the benchmark economic controls and country and year fixed effects. The fuel exclusion uses the common full-basket/non-fuel sample and retains the original full-basket controls. Trade thresholds apply to average annual gross bilateral trade during the measurement window. Country-pair-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

References

Aghion, P., Antràs, P., and Helpman, E. (2007). Negotiating free trade. *Journal of International Economics*, 73(1):1–30.

Bagwell, K. and Staiger, R. (1999). An economic theory of gatt. *American Economic Review*, 89(1):215–248.

Bagwell, K. and Staiger, R. (2002). Economic theory and the interpretation of gatt/wto. *The American Economist*, 46(2):3–19.

Bagwell, K., Staiger, R., and Yurukoglu, A. (2020a). Multilateral trade bargaining: A first look at the gatt bargaining records. *American Economic Journal: Applied Economics*, 12(3):72–105.

Bagwell, K. and Staiger, R. W. (2010a). Backward stealing and forward manipulation in the wto. *Journal of International Economics*, 82(1):49–62.

Bagwell, K. and Staiger, R. W. (2010b). The world trade organization: Theory and practice. *Annual Review of Economics*, 2(Volume 2, 2010):223–256.

Bagwell, K., Staiger, R. W., and Yurukoglu, A. (2020b). "nash-in-nash" tariff bargaining. *Journal of International Economics*, 122:103263.

Baier, S. L. and Bergstrand, J. H. (2004). Economic determinants of free trade agreements. *Journal of International Economics*, 64(1):29–63.

Baldwin, R. and Jaimovich, D. (2012). Are free trade agreements contagious? *Journal of International Economics*, 88(1):1–16.

Becker, G. S. (1973). A theory of marriage: Part I. *Journal of Political Economy*, 81(4):813–846.

Bhagwati, J. (1992). The world trading system at risk. *Princeton University Press*.

Blanchard, E. J. (2007). Foreign direct investment, endogenous tariffs, and preferential trade agreements. *The B.E. Journal of Economic Analysis & Policy*, 7(1):1–52.

Blanchard, E. J., Bown, C. P., and Johnson, R. C. (2025). Global value chains and trade policy. *The Review of Economic Studies*, page rdaf017.

Blanga-Gubbay, M., Conconi, P., and Parenti, M. (2024). Lobbying for globalisation. *The Economic Journal*, 135(666):487–518.

Bloch, F. and Jackson, M. O. (2007). The formation of networks with transfers among players. *Journal of Economic Theory*, 133(1):83–110.

Bulow, J. I., Geanakoplos, J. D., and Klemperer, P. D. (1985). Multimarket oligopoly: Strategic substitutes and complements. *Journal of Political Economy*, 93(3):488–511.

Buzard, K. (2017). Self-enforcing trade agreements and lobbying. *Journal of International Economics*, 108:226–242.

Coase, R. H. (1960). The problem of social cost. *Journal of Law and Economics*, 3:1–44.

Cole, M. T., Lake, J., and Zissimos, B. (2021). Contesting an international trade agreement. *Journal of International Economics*, 128:103410.

Conconi, P., Facchini, G., and Zanardi, M. (2012). Fast-track authority and international trade negotiations. *American Economic Journal: Economic Policy*, 4(3):146–189.

Conconi, P. and Perroni, C. (2002). Issue linkage and issue tie-in in multilateral negotiations. *Journal of international Economics*, 57(2):423–447.

Conconi, P. and Perroni, C. (2009). Do credible domestic institutions promote credible international agreements? *Journal of International Economics*, 79(1):160–170.

Conconi, P. and Perroni, C. (2012). Conditional versus unconditional trade concessions for developing countries. *Canadian Journal of Economics*, 45(2):613–631.

Conconi, P. and Sahuguet, N. (2009). Policymakers' horizon and the sustainability of international cooperation. *Journal of Public Economics*, 93(3-4):549–558.

de Fontenay, C. C. and Gans, J. S. (2014). Bilateral bargaining with externalities. *Journal of Industrial Economics*, 62(4):756–788.

Egger, P. and Larch, M. (2008). Interdependent preferential trade agreement memberships: An empirical analysis. *Journal of International Economics*, 76(2):384–399.

Ellingsen, T. and Paltseva, E. (2016). Confining the coase theorem: contracting, ownership, and free-riding. *The Review of Economic Studies*, 83(2):547–586.

Facchini, G., Silva, P., and Willmann, G. (2013). The customs union issue: Why do we observe so few of them? *Journal of International Economics*, 90(1):136–147.

Facchini, G., Silva, P., and Willmann, G. (2021). The political economy of preferential trade agreements: An empirical investigation. *The Economic Journal*, 131(640):3207–3240.

Furusawa, T. and Wen, Q. (2002). Disagreement points in trade negotiations. *Journal of International Economics*, 57(1):133–150.

Grossman, G. M. (2016). The Purpose of Trade Agreements. NBER Working Papers 22070, National Bureau of Economic Research, Inc.

Grossman, G. M. and Helpman, E. (1994). Protection for sale. *The American Economic Review*, 84(4):833–850.

Grossman, G. M. and Helpman, E. (1995). The politics of free-trade agreements. *American Economic Review*, 85(4):667–690.

Harstad, B. (2007). Harmonization and side payments in political cooperation. *American Economic Review*, 97(3):871–889.

Harstad, B. (2023). Pledge-and-review bargaining. *Journal of Economic Theory*, 207:105574.

Harstad, B. (2026). Contingent trade agreements: Linking free trade and environmental conservation. *The Economic Journal*. Forthcoming.

Horn, H., Maggi, G., and Staiger, R. W. (2010). Trade agreements as endogenously incomplete contracts. *American Economic Review*, 100(1):394–419.

Jackson, M. O. and Wilkie, S. (2005). Endogenous games and mechanisms: Side payments among players. *The Review of Economic Studies*, 72(2):543–566.

Johnson, H. G. (1953). Optimum tariffs and retaliation. *The Review of Economic Studies*, 21(2):142–153.

Karacaovali, B. and Limão, N. (2008). The clash of liberalizations: Preferential vs. multilateral trade liberalization in the european union. *Journal of International Economics*, 74(2):299–327.

Lake, J. (2017). Free trade agreements as dynamic farsighted networks. *Economic Inquiry*, 55(1):31–50.

Limão, N. (2007). Are preferential trade agreements with non-trade objectives a stumbling block for multilateral liberalization? *The Review of Economic Studies*, 74(3):821–855.

Limão, N. (2006). Preferential Trade Agreements as Stumbling Blocks for Multilateral Trade Liberalization: Evidence for the United States. *American Economic Review*, 96(3):896–914.

Limão, N. (2016). Preferential Trade Agreements. NBER Working Papers 22138, National Bureau of Economic Research, Inc.

Limão, N. and Maggi, G. (2015). Uncertainty and trade agreements. *American Economic Journal: Microeconomics*, 7(4):1–42.

Magee, C. S. (2003). Endogenous preferential trade agreements: An empirical analysis. *Contributions in Economic Analysis & Policy*, 2(1):1–17.

Maggi, G. (2014). International Trade Agreements. In Gopinath, G., Helpman, ., and Rogoff, K., editors, *Handbook of International Economics*, volume 4 of *Handbook of International Economics*, chapter 0, pages 317–390. Elsevier.

Maggi, G. (2016). Chapter 9 - issue linkage. volume 1 of *Handbook of Commercial Policy*, pages 513–564. North-Holland.

Maggi, G. and Rodriguez-Clare, A. (1998). The value of trade agreements in the presence of political pressures. *Journal of Political Economy*, 106(3):574–601.

Maggi, G. and Rodríguez-Clare, A. (2007). A political-economy theory of trade agreements. *American Economic Review*, 97(4):1374–1406.

Mansfield, E. D., Milner, H. V., and Rosendorff, B. P. (2002). Why democracies cooperate more: Electoral control and international trade agreements. *International Organization*, 56(3):477–513.

Milgrom, P. and Roberts, J. (1990). Rationalizability, learning, and equilibrium in games with strategic complementarities. *Econometrica*, 58(6):1255–1277.

Milgrom, P. and Shannon, C. (1994). Monotone comparative statics. *Econometrica*, 62(1):157–180.

Mitra, D. (2002). Endogenous political organization and the value of trade agreements. *Journal of International Economics*, 57(2):473–485.

Park, J.-H. (2011). Enforcing international trade agreements with imperfect private monitoring. *The Review of Economic Studies*, 78(3):1102–1134.

Topkis, D. M. (1998). *Supermodularity and Complementarity*. Princeton University Press, Princeton, NJ.

Watson, J. (2024). Contractual chains. *Econometrica*. Open access article; John Wiley & Sons Ltd on behalf of The Econometric Society.

Yamada, A. (2003). Efficient equilibrium side contracts. *Economics Bulletin*, 3(6):1–7.